Current release baseline

22 October 2026

Scope. This revision governs the current local system. The existing requirement and screen identifiers are retained. Earlier diagrams, endpoint sketches and routing algorithms are historical or deferred wherever they conflict with this baseline.

Navigation. External Google Maps URLs are the selected no-key approach. A user click sends only a published mosque entrance and walking or two-wheeler mode. No embedded map, Google API billing, visitor origin, private contact or route-duration response is involved. Directions work without a scheduled session. The app makes no arrival-time or reachability calculation.

Accounts and trust. Only Admin issues focal credentials; only the responsible focal person issues representative credentials. Contact details are confirmed independently by the issuer. Temporary passwords expire after 24 hours and require a first-login change. No SMS OTP or verified-email recovery is required. Administrator MFA uses free local TOTP and single-use recovery codes. Imam phone-call evidence remains a separate timetable-verification process.

Implementation. React/TypeScript frontend, Express API and a separate worker use persistent MongoDB. Authorization, revisions, audit and outbox writes are enforced on the server. Public responses exclude private contacts. Local ports are 5173 for the frontend and 4000 for the API; production hosting is not provisioned.

Acceptance boundaries. The core local flows have automated and mobile-viewport browser coverage. Large-scale pagination, full Urdu copy review, combined registration resubmission, production recovery/load acceptance and physical old-Android validation remain outstanding. Google API routing and forecasts are deferred. See Documents/CURRENT_SYSTEM.md and backend/docs/IMPLEMENTATION_REPORT.md for the maintained implementation record.

DiscoverSalah
Design and Test Specification

Phase 3

Implementation baseline and release evidence plan

Document control	Value
Document ID	DS-DTS-P3-001
Version and date	1.1 • 15 October 2026
Product Owner	Muhammad Ahmad
Baseline	Proposal v4.0; Phase 1 SRS v1.2; Phase 2 SDS v1.1
Purpose	Turn the agreed design and 15 core prototypes into implementable contracts, executable test cases and pilot release gates.
Evidence status	15 HTML prototypes reviewed. Production implementation, integration tests, field calibration and independent validation remain unverified.
Initial release	Free English/Urdu website, starting with any management-approved mosque with assigned maintainers. Walking and motorcycle support depend on validated route evidence.

The product must help residents and travelers find a nearby mosque and its actual Jama’at time without implying that an estimated arrival or an unverified timetable is reliable. Local Representatives submit information; authorized Focal Persons verify it by calling the registered Imam and approve the exact submitted version.

This specification is ready for implementation planning. Release approval depends on the evidence and unresolved decisions recorded in Sections 6 and 7; the UI prototypes alone do not establish a working service.

Contents

Abstract 3
1 Introduction 4
1.5 Business goals and conventions 5
2 Technical architecture 6
2.1 Application and data architecture 7
2.1.1 Schedule data model 8
2.1.2 Identity and operational records 9
2.2 Component interactions and collaborations 10
2.2.1 Verification and publication 11
2.3 Design reuse and design patterns 12
3 Detailed component design 13
3.1.1 Session and reachability decisions 14
3.2 External entities and human interfaces 15
3.4 Maps and bilingual behavior 16
4 Screenshots and prototype 17
4.2.1 UI01 Home and mosque discovery 18
4.2.2 UI02 Choose your location 19
4.2.3 UI03 Search results and available Jamaats 20
4.2.4 UI04 Map results and nearby mosques 21
4.2.5 UI05 Mosque details and timetable 22
4.2.6 UI06 Management sign in 23
4.2.7 UI07 Local Representative Dashboard 24
4.2.8 UI08 Mosque Management Overview 25
4.2.9 UI09 Timetable Editor 26
4.2.10 UI10 Review and Submit Timing Changes 27
4.2.11 UI11 Submission Details and Status 28
4.2.12 UI12 Focal Person Dashboard and Review Queue 29
4.2.13 UI13 Timing Verification and Approval 30
4.2.14 UI14 Mosque Registration 31
4.2.15 UI15 Admin Dashboard and Assignment Management 32
4.3 Prototype corrections and supporting coverage 33
4.4 Additional information 34
5 Other design details 35
5.1.1 Privacy retention and account recovery 36
5.2 Reliability and operations 37
6 Test specification and results 38
6.2 Summary of test results 59
6.3 Release acceptance gates 60
7 Revised project plan 61
7.3 Dependencies and delivery controls 62
7.4 Accepted launch decisions 63
References 64
Appendix A Glossary 65
Appendix B Independent verification and validation 66

The structure follows the supplied Phase 3 template. Screen figures are selected readable crops, not full scrolling-page miniatures. Page numbers refer to this edition.

Abstract

DiscoverSalah is a community-maintained mosque and congregation discovery website for people at home and while traveling. General prayer calculations do not establish a particular mosque's congregation time. The service therefore records mosque-specific schedules, approved changes, verification provenance and practical entrance information. A conservative arrival model considers walking or motorcycle travel, preparation and uncertainty; Whole City search supports longer Friday and Eid journeys.

The application is designed as a React web client, a Node.js/Express modular API, a background worker and MongoDB. Immutable proposals, independent approval, scoped assignments and durable review notifications protect timetable integrity. Registered Imam contact information supports remote verification and remains private. The first release is free, with maintenance responsibilities established from launch. Later funding discussions with Islamic organizations do not introduce subscriptions, donations or payroll into the present build.

This phase specifies architecture, interfaces, detailed decision rules, the 15 supplied core UI pages, correction priorities, test cases and a conditional delivery plan. Prototype inspection is complete for documentation purposes; software execution results, routing calibration and independent review require later evidence. The intended outcome is a controlled pilot whose public claims match the quality of its data and operating capacity.

Evidence and change control

Requirements take precedence over illustrative screen text. Conflicting prototype text is logged in Section 4.3 and must be corrected before acceptance. Parameters explicitly labeled proposed require Product Owner approval and field evidence. New scope must update the SRS, this specification and corresponding tests together.

Version	Change	Approval
1.0	Initial specification, screen review and test plan.	Initial edition.
1.1	Accepted launch, language, maps, privacy and recovery policies; four additional tests.	Owner direction 15 Oct 2026; release evidence pending.

1 Introduction

1.1 Product

DiscoverSalah is a custom browser-based information service. Anonymous visitors select an origin, prayer and date, then compare participating mosques and congregation sessions. Management users maintain those records through controlled submissions and verification. It is not a prayer-calculation authority, a transport guarantee or a mosque attendance system.

1.2 Background

Prayer-time calculators address astronomical prayer windows; map directories address places and navigation; local notices and messaging groups communicate mosque-specific changes. DiscoverSalah combines location discovery with accountable local timetable maintenance. Its proposed distinction is the verification and publication workflow, not an unproven claim of better coverage or accuracy than other products. No institutional previous-project catalogue or comparative field study has been established for this phase; neither is presented as evidence.

1.3 Objectives

Objective	Acceptance evidence
Find relevant congregation sessions	Location, selected prayer/date, Nearby and Whole City scenarios pass; canceled sessions do not reappear.
Keep published information accountable	Independent exact-version approval, private registered contact, audit and concurrency tests pass.
Provide conservative arrival estimates	Supported routes and validated mode-specific uncertainty; timetable-only fallback when evidence fails.
Operate a maintainable pilot	Named coverage and backup owners, stale-record review, restore rehearsal and incident handling demonstrated.
Make the service usable	Core tasks tested on mobile and desktop, keyboard and zoom; problems corrected before release.

1.4 Scope

Included: location selection, list/map discovery, venue details, recurring and exceptional sessions, registration, roles and assignments, draft/review/publication, durable notifications, issue reports, missing-mosque suggestions, contributor applications and moderated expiring notices. The 15 core pages are the initial design set; supporting forms and error states still require implementation design.

Excluded: subscriptions, payments, payroll, advertisements, native mobile apps, automatic Imam calling, automated religious judgments, car or bicycle reachability, OCR timetable extraction and multi-stop journey planning. English and Urdu switching is included. Automated user reminders and future-departure planning are deferred; required in-account submission and decision notifications remain included.

1.5 Business goals and conventions

Business goals

Begin with any willing mosque with management permission, assigned data maintenance and an independent Focal Person; do not wait for three complete areas. Keep public discovery free. Grow through Imam announcements at Friday prayer and mosque notice boards or QR links. Monitor maintenance workload and verification delays so expansion does not outpace local support. Funding discussions with Islamic organizations may follow demonstrated coverage and costs; no funding or payment to contributors is assumed.

Document conventions

Convention	Meaning
Shall	Required behavior for the agreed release scope.
Proposed	A parameter or implementation choice awaiting the stated decision or evidence.
UI01–UI15	Core screen identifiers, matching the user's design sequence.
P3-T01 onward	Phase 3 test identifiers; local identifiers do not replace SRS story IDs.
Not run	No application execution evidence; never a pass or zero-defect claim.
Prototype finding	A source/design issue observed in supplied HTML, separate from an executed software defect.
Times	Mosque wall-clock time in Asia/Karachi; API instants use explicit offsets or UTC.
Typography	11 pt body, consistent heading levels and 10 pt tables; figures sized for reading.

1.6 Miscellaneous

Product Owner, engineering, testing and local operations responsibilities must be assigned before delivery dates are committed. Imam phone calls are human actions, with contact permission and concise verification notes; the system does not record calls by default. Demonstration names, counts, contacts and dates in the supplied UI are fixtures and must not be migrated into production.

The owner's 22 October 2026 decision replaces the earlier Google Maps/Routes API selection with external Google Maps URLs for the current local release. No route API, embedded map, geocoding service, API key or Google billing account is needed. Current credentials and deployment follow the revision page at the start of this document. Earlier route calculations remain deferred design.

2 Technical architecture

The selected design is a custom modular monolith. Online requests handle search, registration, editing and review. A small worker handles durable notifications and scheduled maintenance. Operational counts are limited reporting; no analytical warehouse is needed for the pilot.

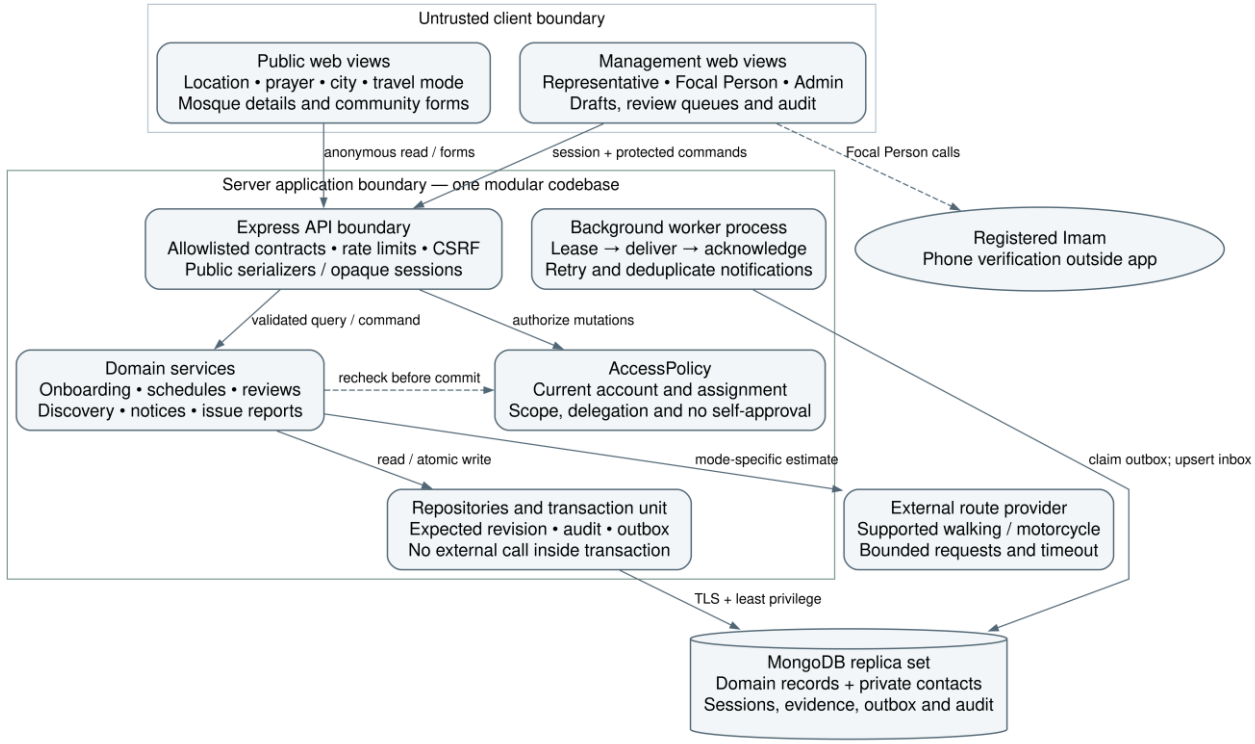


Figure 2.1 Proposed runtime architecture retained from the Phase 2 baseline

The browser communicates with the API over HTTPS. Public reads use safe projections; management writes authenticate and authorize each request. MongoDB transactions commit domain changes, audit and outbox records together. External maps, geocoding and routes remain separate adapters. Hosting and provider selection are open procurement decisions, not deployed infrastructure.

2.1 Application and data architecture

Component	Responsibility	Boundary
React client	Forms, discovery views, accessible feedback and retained query state.	Cannot authorize changes or declare a record verified.
Authentication and scope	Sessions, roles, account and assignment revisions.	Deny by default; check exact target scope on every request.
Venue registry	Entrance, facilities, private Imam contact and onboarding.	Duplicate review precedes publication.
Schedule service	Immutable full proposals, recurring rules and dated exceptions.	One active pending proposal per venue; approved head stays separate.
Review service	Task ownership, call evidence and approve/return/reject.	No self-approval; revision checked at commit.
Discovery service	Session resolution, eligibility, routes and deterministic ordering.	No positive estimate without eligible data and validated route evidence.
Community service	Reports, suggestions, applications and notices.	No anonymous automatic schedule publication.
Worker and operations	Outbox retry, notifications, activation reconciliation and monitoring.	Read-time validity remains authoritative if worker is delayed.

Data responsibilities

Collect only operationally necessary account, assignment, venue, schedule, decision and audit data. Store private contacts separately from public venue fields. Avoid persistent storage of anonymous precise origins; redact location and contact data from logs. Reports and contributor applications use access-controlled records and the retention policy in Section 5.1.1.

2.1.1 Schedule data model

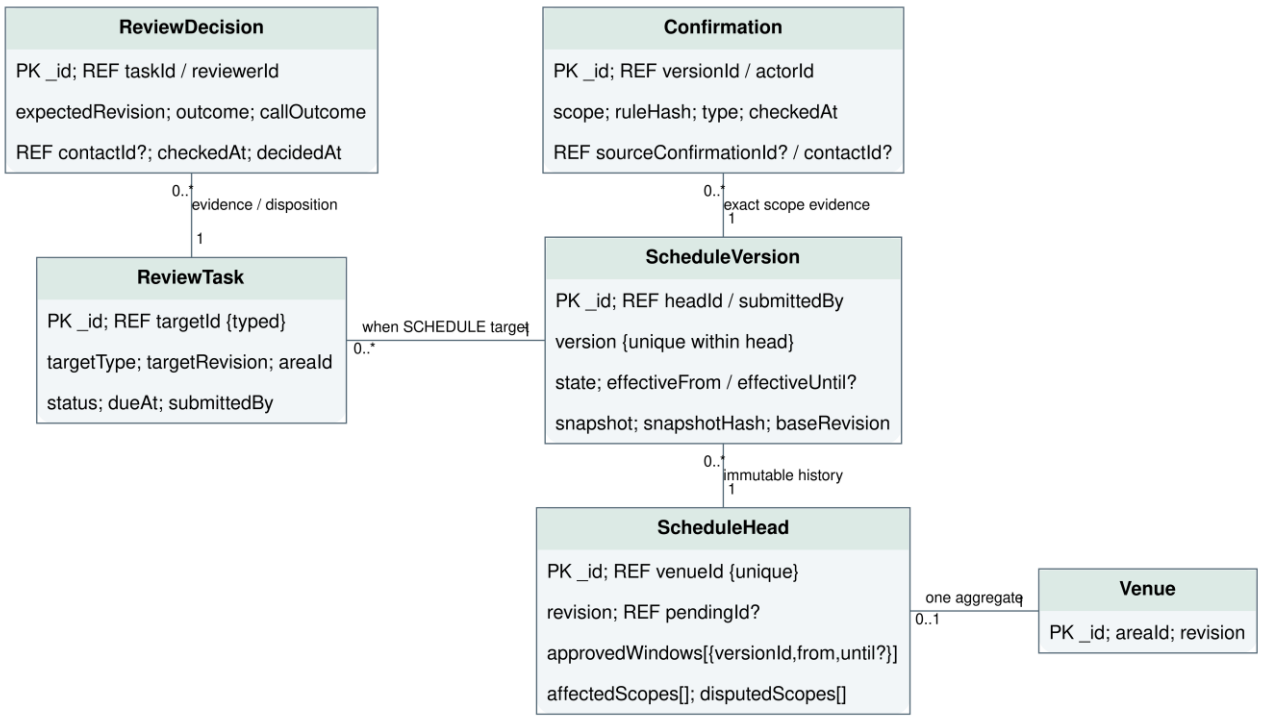


Figure 2.2 Schedule records and relationships retained from Phase 2

A schedule version is an immutable full snapshot. A mutable ScheduleHead identifies approved and pending versions and carries a revision for optimistic concurrency. Explicit cancellations prevent fallback to recurring sessions. MongoDB references require application validation; they are not database-enforced foreign keys.

2.1.2 Identity and operational records

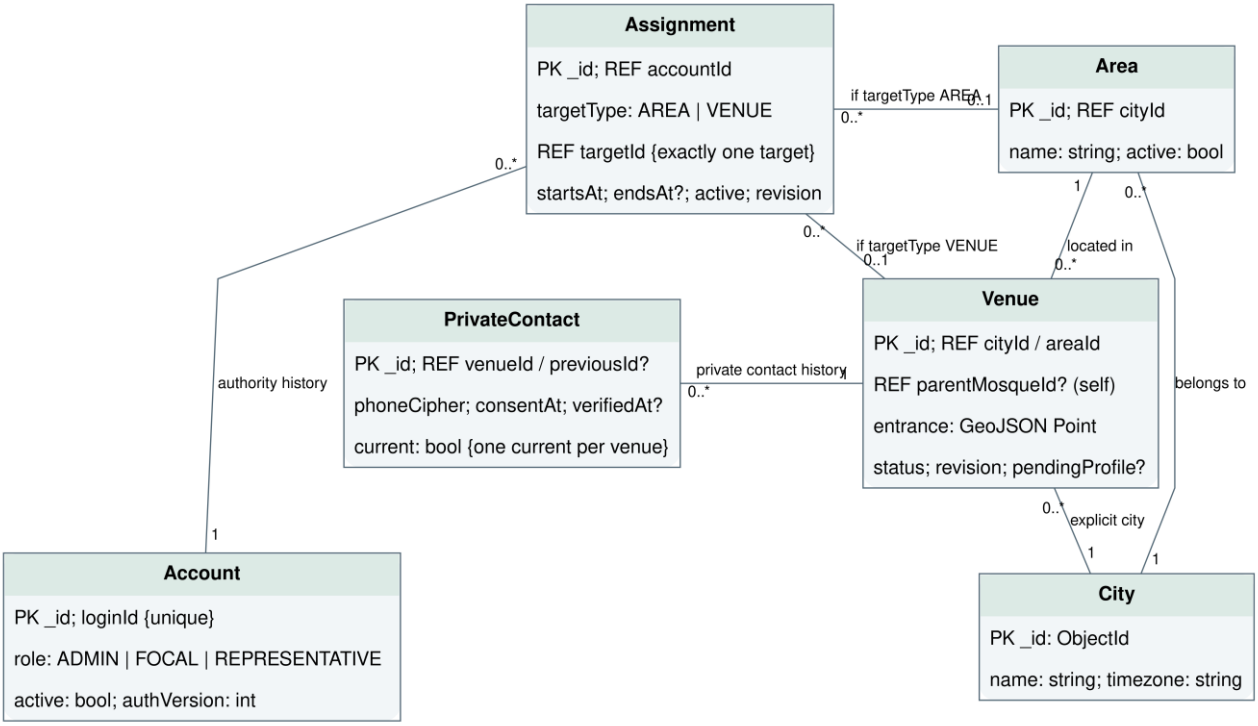


Figure 2.3 Identity, assignments and private venue contact boundary

Record group	Required constraint
Account and assignment	Active account and current scoped assignment; revoked authority fails at commit.
Review task and decision	Unique active task per submitted version; decision includes reviewer, evidence, time and version.
Outbox and notification	Atomic event creation; deduplicate by event and recipient; retry safely.
Audit	Append-only application access; actor, target, request ID and before/after revisions without secrets.

2.2 Component interactions and collaborations

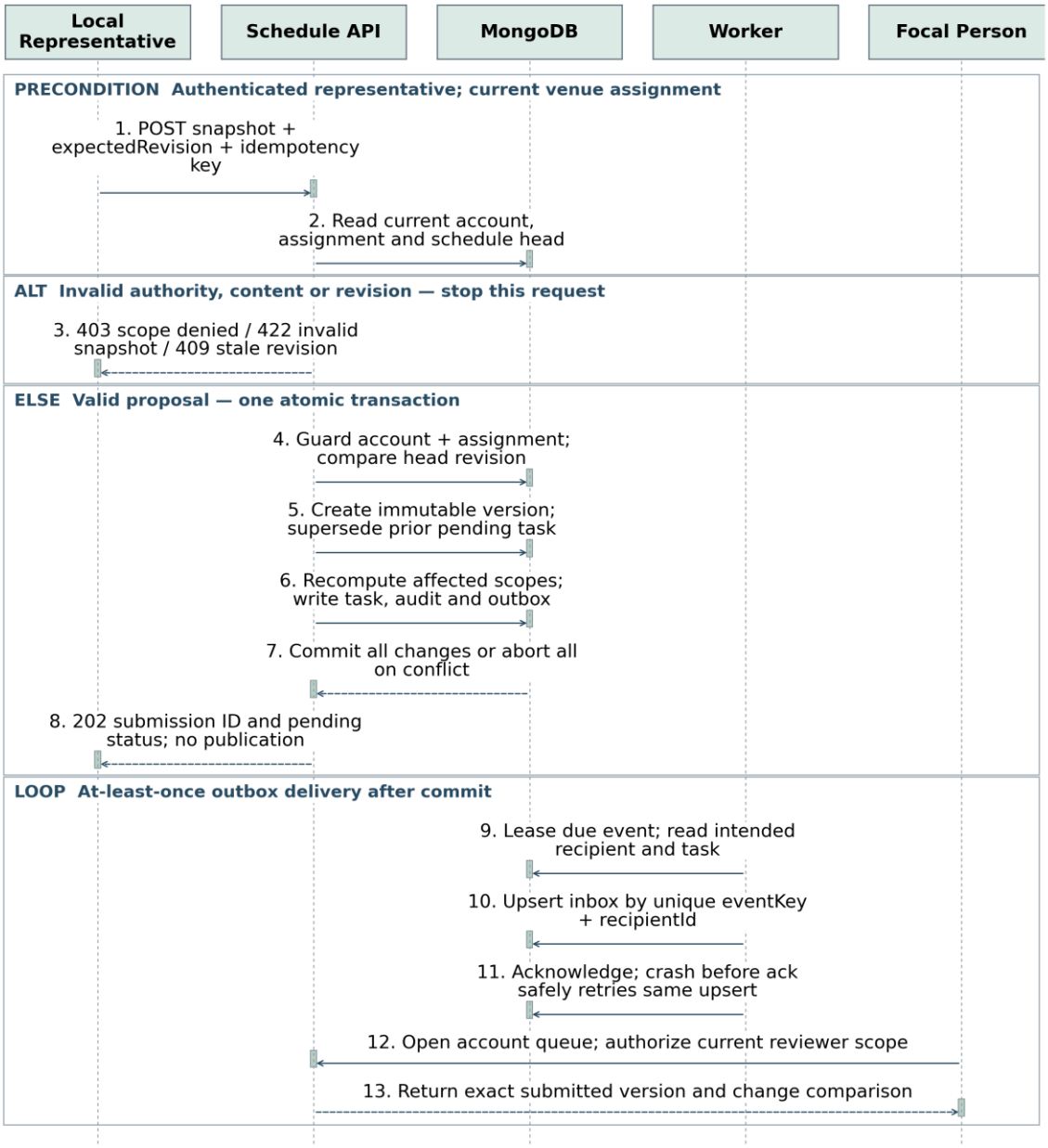


Figure 2.4 Submission transaction and notification handoff

Submission uses an idempotency key and expected head revision. In a single transaction the service checks authority, stores the snapshot, supersedes the prior pending version where applicable, advances the head, creates the review task and writes audit/outbox records. A response lost after commit can be retried without creating another proposal. A changed payload with the same key is rejected.

2.2.1 Verification and publication

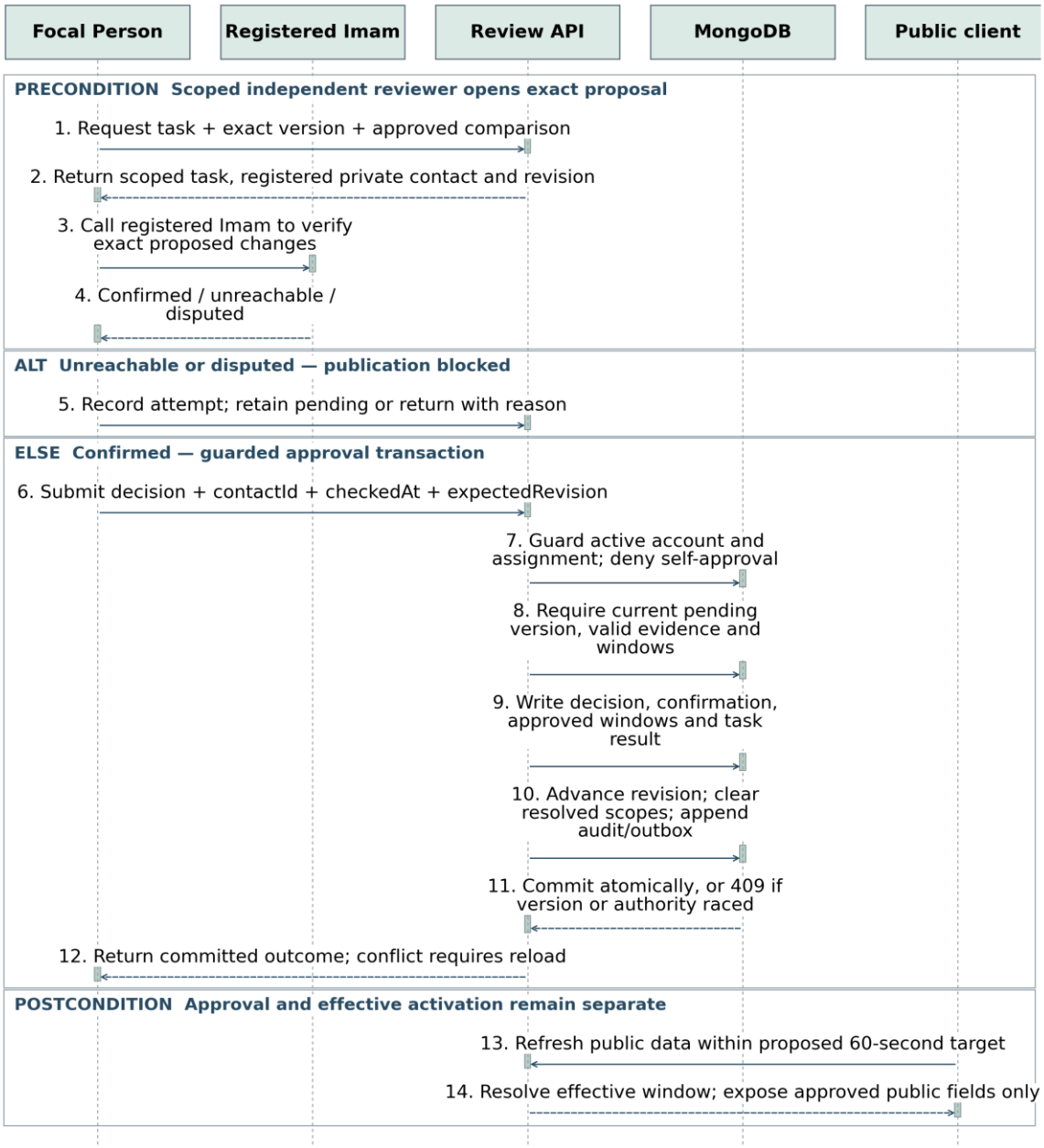


Figure 2.5 Independent verification and exact-version approval

The Focal Person calls the registered Imam; a displayed phone number or ticked checkbox alone is not evidence of a completed call. Record who was contacted, when, the verified scope and outcome. Unreachable or ambiguous contact leaves the task pending or returns it for correction. Future-effective approval does not publish early.

2.3 Design reuse and design patterns

Pattern	Application	Reason
Layered modules	Routes call application services; repositories isolate MongoDB.	Keep domain decisions outside UI handlers.
Adapter	Routing, geocoding, map display and navigation.	Provider changes do not rewrite reachability policy.
Pure policy functions	Session resolution, freshness and arrival classification.	Deterministic unit and boundary testing.
Optimistic concurrency	Expected revisions and conditional writes.	Prevent stale approval, lost edits and revocation races.
Transactional outbox	Write domain event in the same transaction as the decision.	Worker recovery does not lose a committed review notification.
Idempotent commands	Request key plus payload hash and stored result.	Safe retries after unreliable mobile connections.
Allowlisted projections	Explicit public and management response models.	Prevent private phone and account data leakage.

2.4 Technology architecture

Implemented stack: React and TypeScript, Vite, Node.js/Express TypeScript API, native MongoDB driver and a separate worker. The configured persistent MongoDB replica set is connected. Dependencies are recorded in repository lockfiles. The original standalone HTML remains a visual reference.

Current deployment is local: frontend http://127.0.0.1:5173, API on loopback port 4000 and a separately running notification/retention worker. The frontend proxies /api to the backend. MongoDB uses the private backend environment configuration. No Vercel or other production hosting has been provisioned. Production HTTPS, supervision, backup automation and restore acceptance remain future work.

2.5 Architecture evaluation

A modular monolith reduces deployment and operational overhead compared with microservices. MongoDB fits document-shaped immutable proposals but requires deliberate indexes, transactions and integrity checks; PostgreSQL is a credible alternative with stronger relational constraints. Retain the agreed database unless an implementation review establishes a concrete benefit from changing it. A worker and outbox avoid the initial cost of a broker while preserving durable handoff.

Use no-key Google Maps URLs for external directions. The URL contains a validated published entrance and travelmode=walking or travelmode=two-wheeler. Omit the visitor origin and private contacts. Open only after a user click with referrer/opener protection. Do not call a Google route API, infer travel duration or substitute a car route for motorcycle travel.

3 Detailed component design

3.1 Component to component interfaces

Operation	Input and success	Failures and invariants
Discover sessions	POST /api/v1/discover/query: origin, city, prayer, date, scope, filters, cursor. 200: sessions, trust state, server time.	Validate origin/date; preserve prayer/date; explicit coverage and pagination.
Evaluate arrivals	POST /api/v1/discover/evaluate: session IDs/revisions, mode, departure. 200: status, reason, route time and allowances.	Timeout/provider failure => unavailable; final revision check before positive result.
Submit proposal	POST /api/v1/management/venues/{id}/proposals: snapshot, reason, expectedRevision; idempotency key. 202: version/task IDs.	401/403 unauthorized; 409 stale revision; 422 invalid schedule. No partial commit.
Review proposal	POST /api/v1/management/proposals/{id}/decisions: approve/return/reject, evidence, expectedRevision.	403 self/out-of-scope; 409 superseded; 422 incomplete verification.
Change assignment	POST /api/v1/admin/assignments: scope, assignee, reason, expectedRevision.	Check admin scope and conflicts; immediately revoke prior writes and reroute tasks.
Read notification	GET /api/v1/management/notifications; acknowledge through authenticated write.	Owner-only; task exists independently of notification delivery.

Routes above are proposed Phase 3 contracts within the Phase 2 API namespace. Freeze exact request/response schemas in OpenAPI before parallel client/server development. Errors use code, safe message, field errors where relevant and requestId. Never return stack traces or private contacts in public errors.

Validation and transactions

Validate IDs, IANA timezone, calendar dates, wall-clock times, effective intervals, session uniqueness, venue ownership and explicit cancellation. The server derives actor and role from the session. Recheck account/assignment revisions and schedule head inside the write transaction. A UI-disabled button is not an authorization control.

3.1.1 Session and reachability decisions

Resolve an exact prayer/date/session before routing. Apply valid dated overrides and cancellations before recurrence. Multiple Friday sessions are separate choices; ordinary Dhuhr is not silently added unless explicitly confirmed. Eid dates may remain provisional and an outdoor venue requires its own entrance. Khutbah time is separate from congregation start.

Condition in precedence order	Public behavior
Missing, expired, provisional, stale, disputed, affected pending or unmaintained session	Show appropriate timing status; reachability unavailable, with reason.
Unsupported route mode, route failure or unvalidated uncertainty	Timetable remains visible; arrival estimate unavailable.
Raw arrival at or after session start	Unlikely; never suggest speeding.
Raw arrival before start but full allowances do not fit	Tight; explain insufficient margin.
Arrival including preparation and uncertainty at or before start	Reachable estimate, with time and assumptions; no attendance guarantee.

Margin = session start − departure time − route duration − preparation allowance − uncertainty allowance. Proposed minimum preparation: 5 minutes walking, 8 minutes motorcycle, adjustable upward. Mode-specific uncertainty must be field-validated; zero is not an assumed default. Illustrative test: start 13:30, depart 13:00, route 15, preparation 5, uncertainty 5 gives +5 minutes. These fixture values are not field calibration results.

Refresh active results at least every 60 seconds and on resume using server-aligned time. Rank predictably by eligibility group, freshness, travel time, session time and stable ID. Do not claim city-wide fastest ordering from a partially routed batch. Use stable base pagination by session time and ID, and identify the subset evaluated for travel.

Nearby uses the proposed default 5 km and maximum 7 km; Whole City uses participating city membership without the 7 km cap. An origin outside that city may still be used. The city’s geographic scope does not imply every mosque has been onboarded.

3.2 External entities and human interfaces

Interface	Contract and fallback
Map and geocoding providers	Coordinate confidence and selected entrance; disclose attribution. Denied location permission offers manual origin entry. Tile failure preserves list results.
Routing provider	Explicit mode, coordinates and duration; server-held credentials, timeout and quota handling. Proposed timeout 8 seconds; no-route result never becomes a zero-minute route.
External navigation	User-initiated link with destination entrance and supported mode. Label as external; do not promise the provider will preserve the app's arrival model.
Imam phone call	Focal Person calls registered private number and records verification evidence. No automatic call, OTP substitute or required physical inspection.
Notification channel	Durable in-app notifications and review queues are implemented through a transactional outbox worker. External email/SMS/push delivery and reminder services are not enabled. Issued credentials are handed over privately; SMS/email invitation and recovery endpoints are retired.

3.3 Human interface rules

Use plain labels: Walking, Motorcycle, Jama'at time, Last phone verification and Representative reconfirmation. Keep approved values distinct from pending proposals. Explain why a result is unavailable. Never use color alone for status or put private Imam contact details on public pages.

The supplied design uses Inter with 16 px body text, 24 px line height, forest green, mint surfaces and white cards. Normalize the production tokens against the earlier design brief before implementation: primary #146B4A, background #F7FAF8, text #172B23, secondary #52635B, warning #8A4B08 and error #B42318. Screens in this document retain the supplied palette. Target 48 px controls, visible keyboard focus, meaningful labels, logical headings, 200% zoom and mobile reflow. Contrast and assistive-technology behavior must be tested; conformance is not yet established.

The public flow must work without an account. Invite-only management sign-in resolves permissions from the server. A contributor application is a request for consideration, not self-assignment. Add confirmation only for consequential actions such as approval, reassignment and revocation, and preserve entered data when validation fails.

3.4 Maps and bilingual behavior

Current public behavior: show timetable and timing-confidence information, then offer external Google Maps directions. Directions remain available even when no session exists for the selected prayer/date. No Estimated reachable or Time is tight claim is calculated in this release. A future route adapter requires separate approval, implementation, cost controls and field validation.

Google documents walking and two-wheeler routes as beta and requires a warning about potentially incomplete paths. Display the applicable provider warning and attribution where routes are shown. Motorcycle calls have higher billing rates. Keep restricted routing credentials on the server and apply provider quotas, application request limits and billing alerts; alerts alone do not enforce a spending cap.

Route a bounded, visibly identified subset of eligible results. Whole City still lists participating matches through stable pagination, including unrouted results labeled not evaluated. Do not equate unevaluated results with unreachable results or claim the fastest result across a partially evaluated city. On provider failure or usage-limit exhaustion retain timetable and map/list discovery with arrival unavailable.

Field-test walking and motorcycle allowances separately before enabling positive estimates. Preserve the preparation and uncertainty model in Section 3.1.1. Navigation opens only on explicit user action and uses the verified entrance; disclose the handoff to the external provider.

English and Urdu launch requirements

Provide a visible language switch on public and management pages. Persist the preference locally; keep the current page, selected prayer/date, origin, filters and unsaved form content. Translate labels, errors, status explanations and core help; do not automatically translate names or alter entered timetable values.

Use Urdu right-to-left layout and an Urdu-capable font. Keep phone numbers, email addresses, URLs and numeric time fields isolated in the correct direction. Test mixed-language content, mobile reflow, keyboard navigation and all 15 core pages in both languages. The supplied screenshots remain English prototype references; Urdu layouts require implementation and acceptance testing.

4 Screenshots and prototype

4.1 Workflow

Journey	Core pages	Completion and failure handling
Discover a congregation	UI01 → UI02 → UI03 or UI04 → UI05	Choose origin and exact prayer/date; inspect trust and route status; navigate only on explicit action. Retain filters between list/map/details.
Maintain timetable	UI06 → UI07 → UI08 → UI09 → UI10 → UI11	Edit draft, review differences, submit once, track immutable version. Conflict returns to reload/reconcile, not silent overwrite.
Verify timings	UI06 → UI12 → UI13 → UI11	Check scoped task, call registered Imam, record evidence and decide exact revision. Unreachable contact cannot produce approval.
Register mosque	UI07 or UI08 → UI14 → review task	Capture entrance, private contact permission and first timetable. Duplicate review and Focal verification precede public activation.
Manage coverage	UI06 → UI15 → UI12 or UI08 as authorized	Assign/replace staff, resolve uncovered scope and reassign queued work. Privileged actions logged; no self-approval bypass.

4.2 Screen figure conventions

All 15 supplied HTML designs are represented below. Eight exported PNG files were failed-fetch placeholders and seven were narrow previews. Figures are therefore rendered from the supplied HTML and cropped to readable task regions. They are prototype illustrations, not application test evidence. Crops focus on selected task regions; visible developer scenario controls are not release features; Section 4.3 records corrections rather than silently treating conflicting display text as approved requirements.

Each page states actor, input/output, next action and acceptance focus. Supporting states such as loading, empty results, permission denial, expired sessions, conflicts and failed submission belong to these pages, not to 90 independently designed pages. Additional workflows still need small forms or focused views as listed after the core set.

4.2.1 UI01 Home and mosque discovery

Actor: Visitor. Select location, prayer, date, walking or motorcycle and Nearby or Whole City.

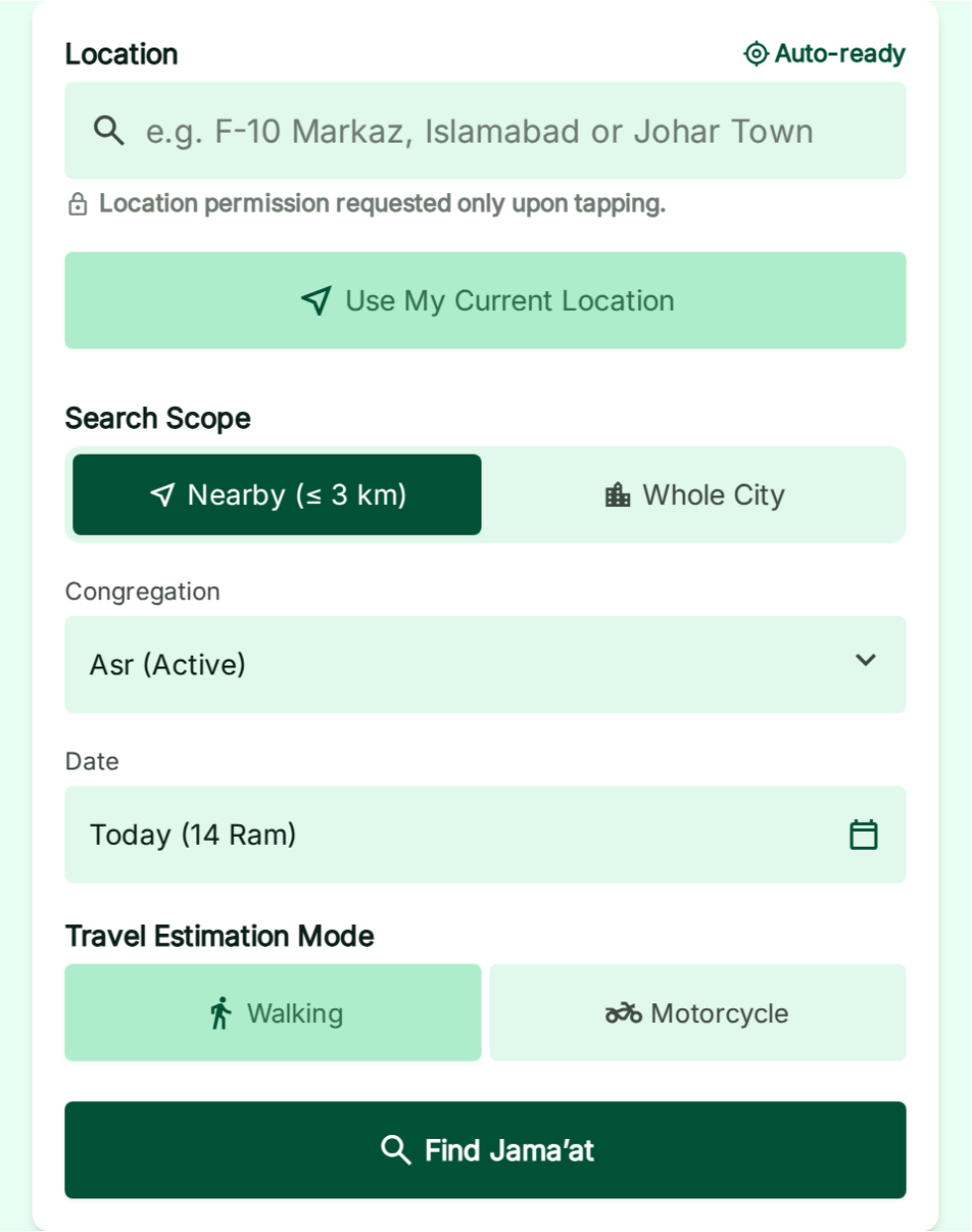


Figure 4.1 UI01 selected task region from the supplied prototype

Flow: Open UI02 for origin selection; show results in UI03. Keep discovery free and explain participating coverage.

Acceptance focus: No location => prompt or manual selection; no invented nearby results. Home metrics and example mosques are demonstration data.

4.2.2 UI02 Choose your location

Actor: Visitor. Use device location, search an address or place a pin; confirm the actual origin.

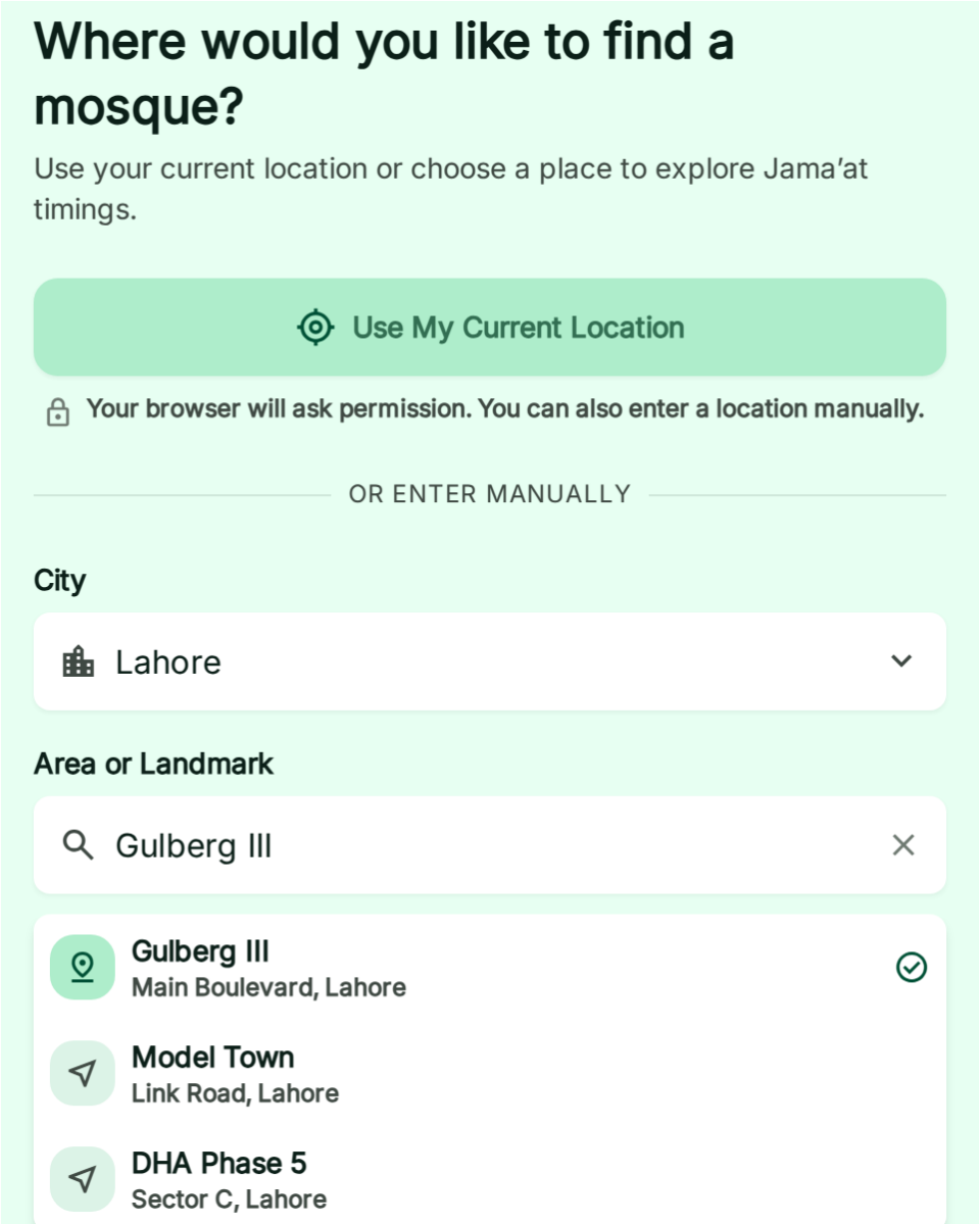


Figure 4.2 UI02 selected task region from the supplied prototype

Flow: Return the selected origin and city to UI01/UI03/UI04 without discarding prayer/date.

Acceptance focus: Permission denied, inaccurate coordinates and out-of-coverage states must remain usable with manual entry.

4.2.3 UI03 Search results and available Jamaats

Actor: Visitor. Compare the selected prayer/date, schedule provenance, distance and arrival status.

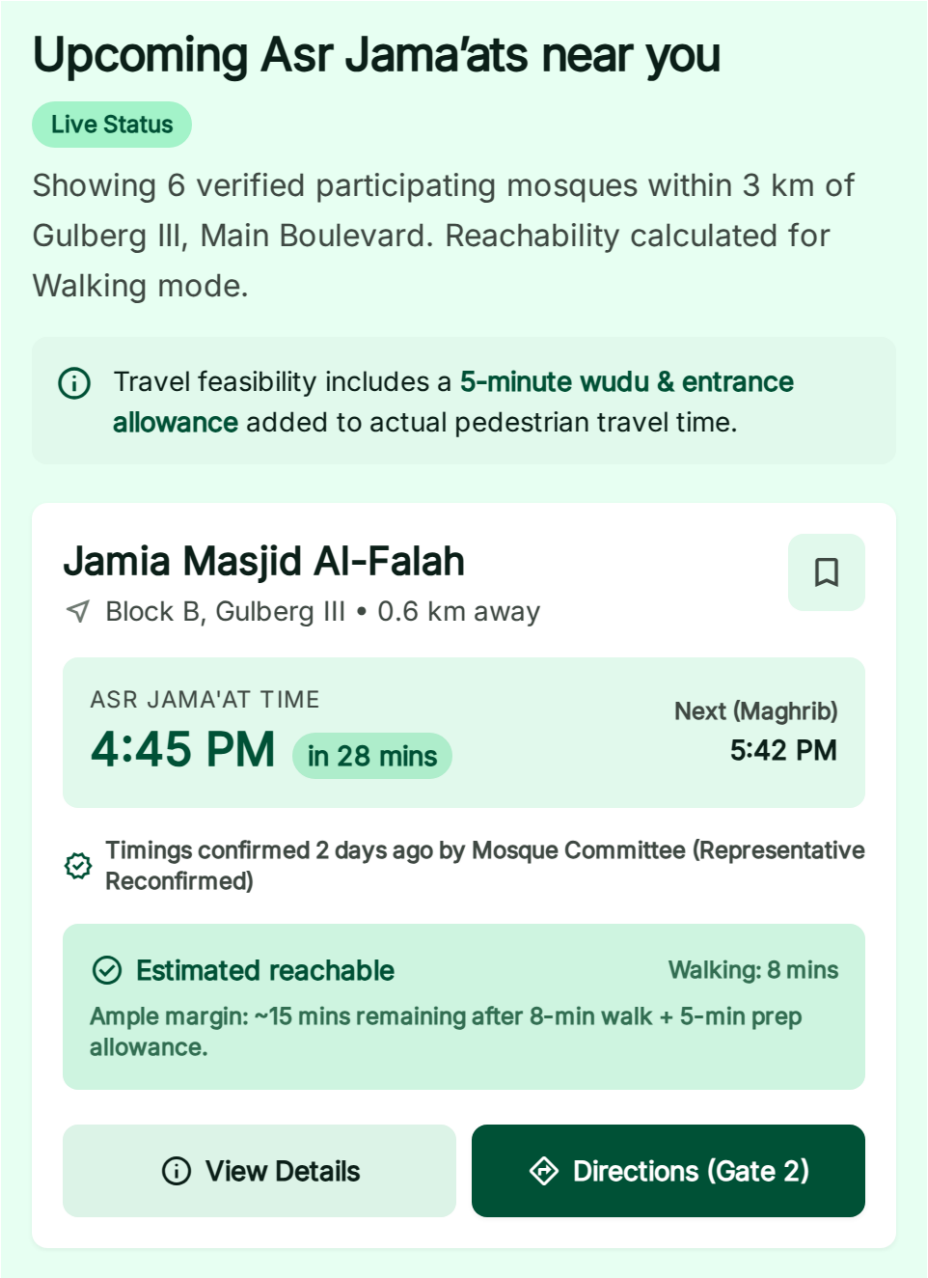


Figure 4.3 UI03 selected task region from the supplied prototype

Flow: Open UI05 for a session or UI04 for the same filtered results.

Acceptance focus: Remove driving advice; use validated per-mode uncertainty and identify partially evaluated city results.

4.2.4 UI04 Map results and nearby mosques

Actor: Visitor. Explore matching venue entrances and select a marker or accessible list equivalent.

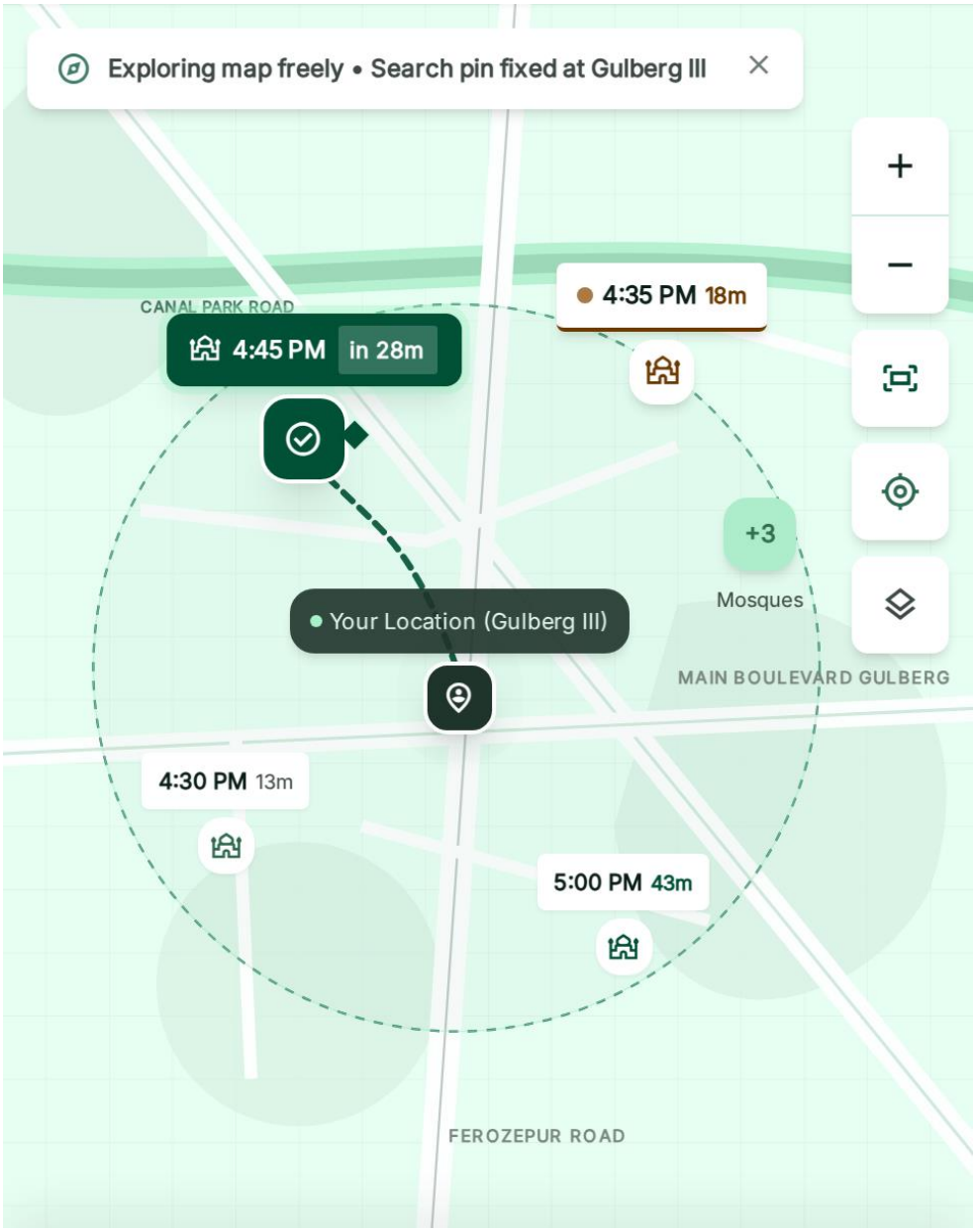


Figure 4.4 UI04 selected task region from the supplied prototype

Flow: Selected marker opens the matching UI05 session; retain query when switching to UI03.

Acceptance focus: Map failure preserves the list; a marker does not itself establish verification or route eligibility.

4.2.5 UI05 Mosque details and timetable

Actor: Visitor. Read approved sessions, last verification, facilities, notices and entrance information.

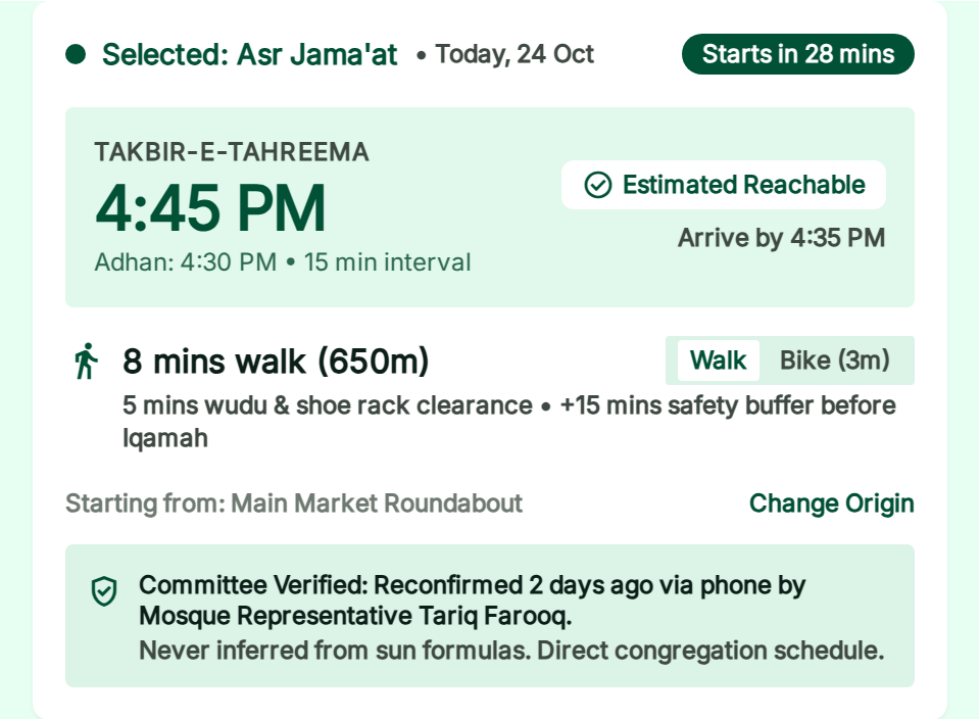


Figure 4.5 UI05 selected task region from the supplied prototype

Flow: Choose exact session and open external navigation deliberately; report errors through supporting form.

Acceptance focus: Separate representative reconfirmation from Focal phone verification. Do not expose private Imam contacts.

4.2.6 UI06 Management sign in

Actor: management account owner. Sign in with the issuer-confirmed contact number and password. Administrators complete TOTP or single-use recovery-code MFA. A temporary password must be changed before management commands become available.

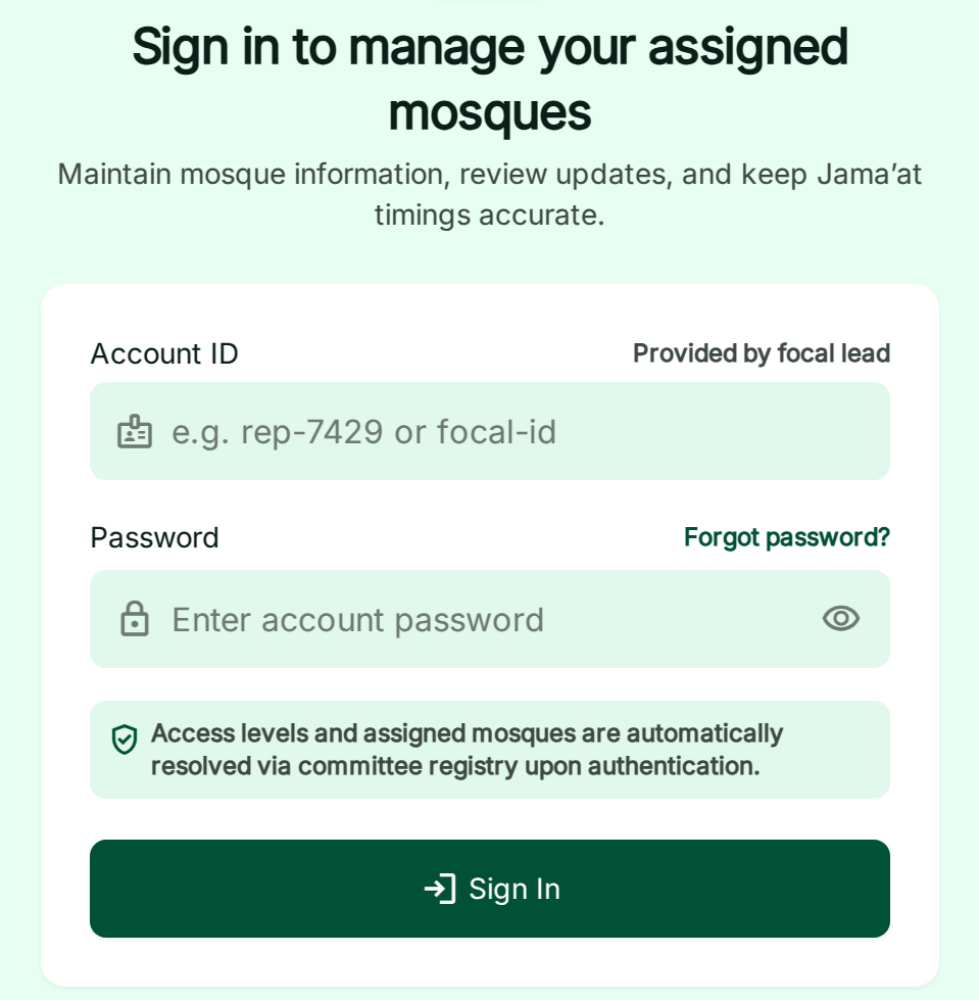


Figure 4.6 UI06 selected task region from the supplied prototype

Flow: Server resolves current assignment and routes to UI07, UI12 or UI15.

Acceptance focus: Correct anonymous timing-update copy. Recovery must have its own verified flow, not the expired-session demo.

4.2.7 UI07 Local Representative Dashboard

Actor: Local Representative. See assigned mosques, pending actions, aging records and recent decisions.

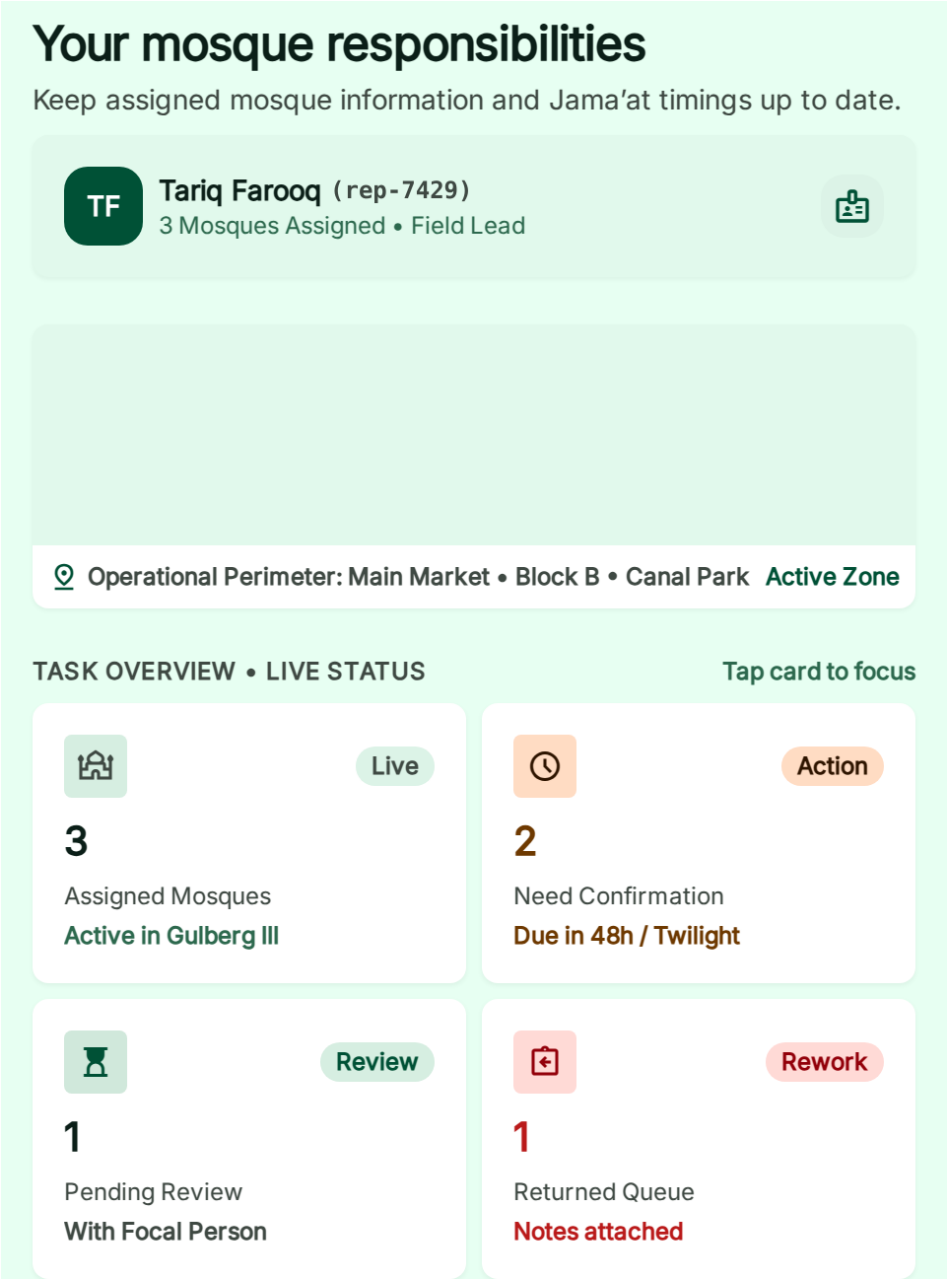


Figure 4.7 UI07 selected task region from the supplied prototype

Flow: Open UI08 for an assigned mosque or UI14 for authorized registration.

Acceptance focus: Unassigned/expired access state shows no write authority. Replace unsupported 30-day expiry claims.

4.2.8 UI08 Mosque Management Overview

Actor: Scoped Local Representative or reviewer. Inspect current approved timetable, pending proposal, verification and venue responsibilities.

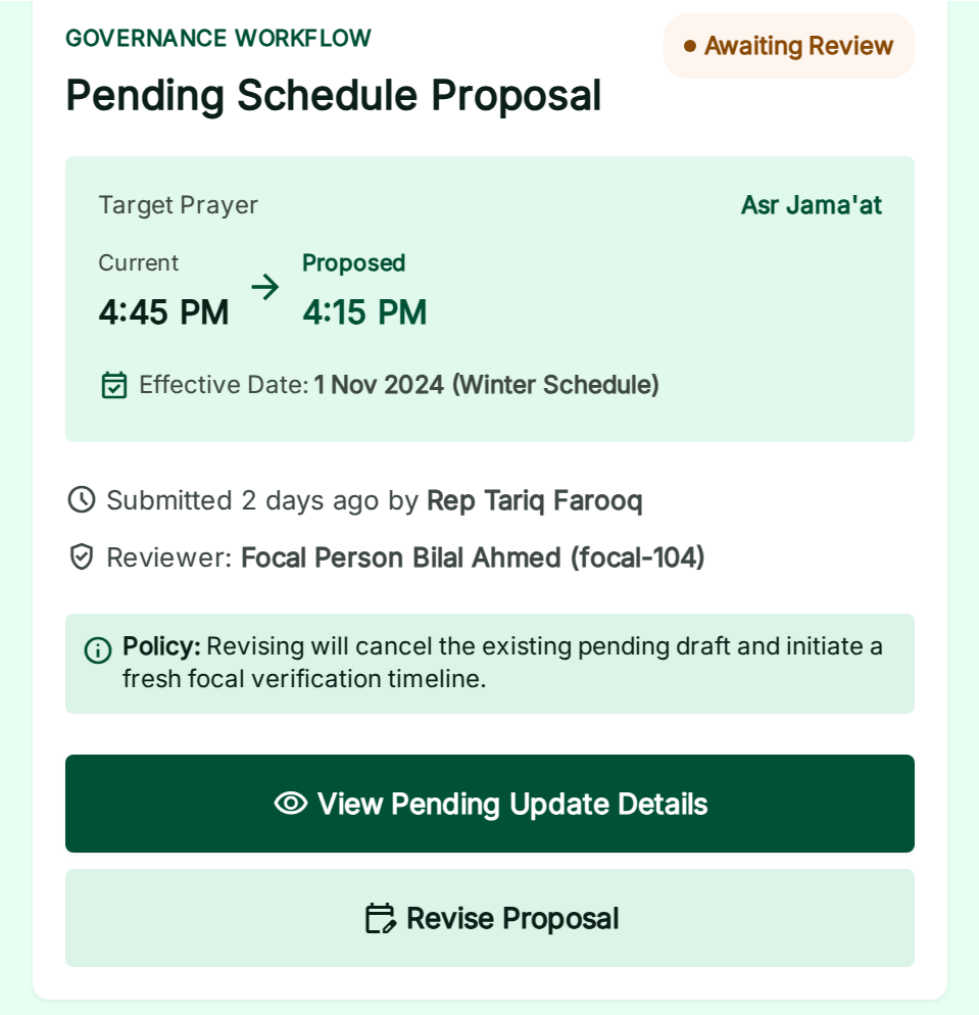


Figure 4.8 UI08 selected task region from the supplied prototype

Flow: Open UI09 to edit; UI11 tracks the current pending version.

Acceptance focus: Current and proposed schedules must remain distinct; private contact panel is management-only.

4.2.9 UI09 Timetable Editor

Actor: Scoped Local Representative. Enter actual mosque congregation times, change reason, dates and explicit exceptions.

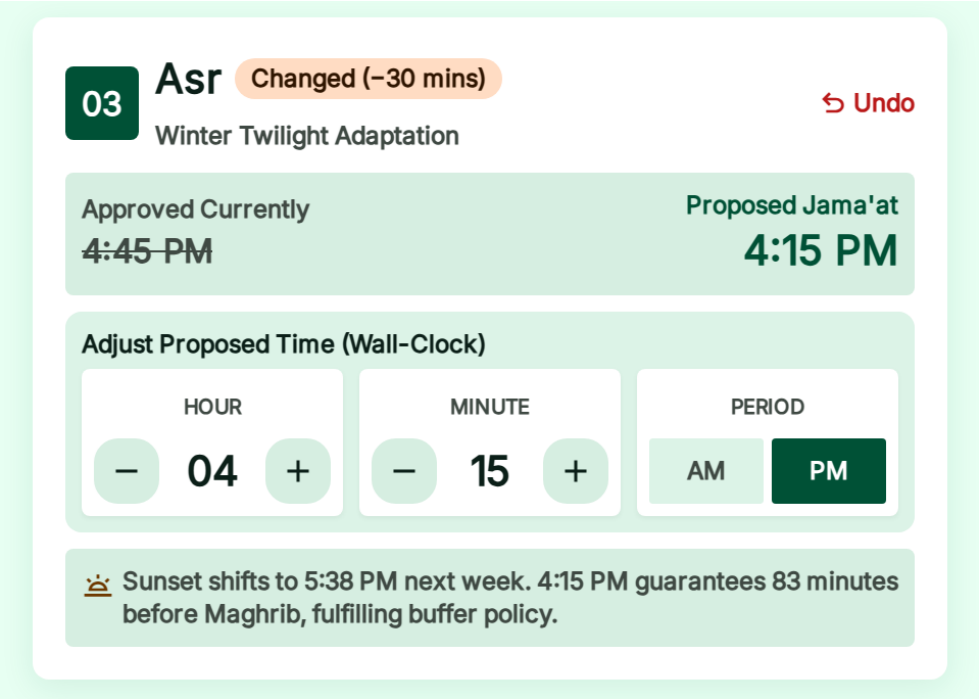


Figure 4.9 UI09 selected task region from the supplied prototype

Flow: Save draft then UI10 review; unsaved changes prompt before leaving.

Acceptance focus: Remove invented twilight/Sunnah rule. Server validates format and conflicts without inventing religious scheduling policy.

4.2.10 UI10 Review and Submit Timing Changes

Actor: Scoped Local Representative. Compare approved and proposed values, effective dates and reason before submission.

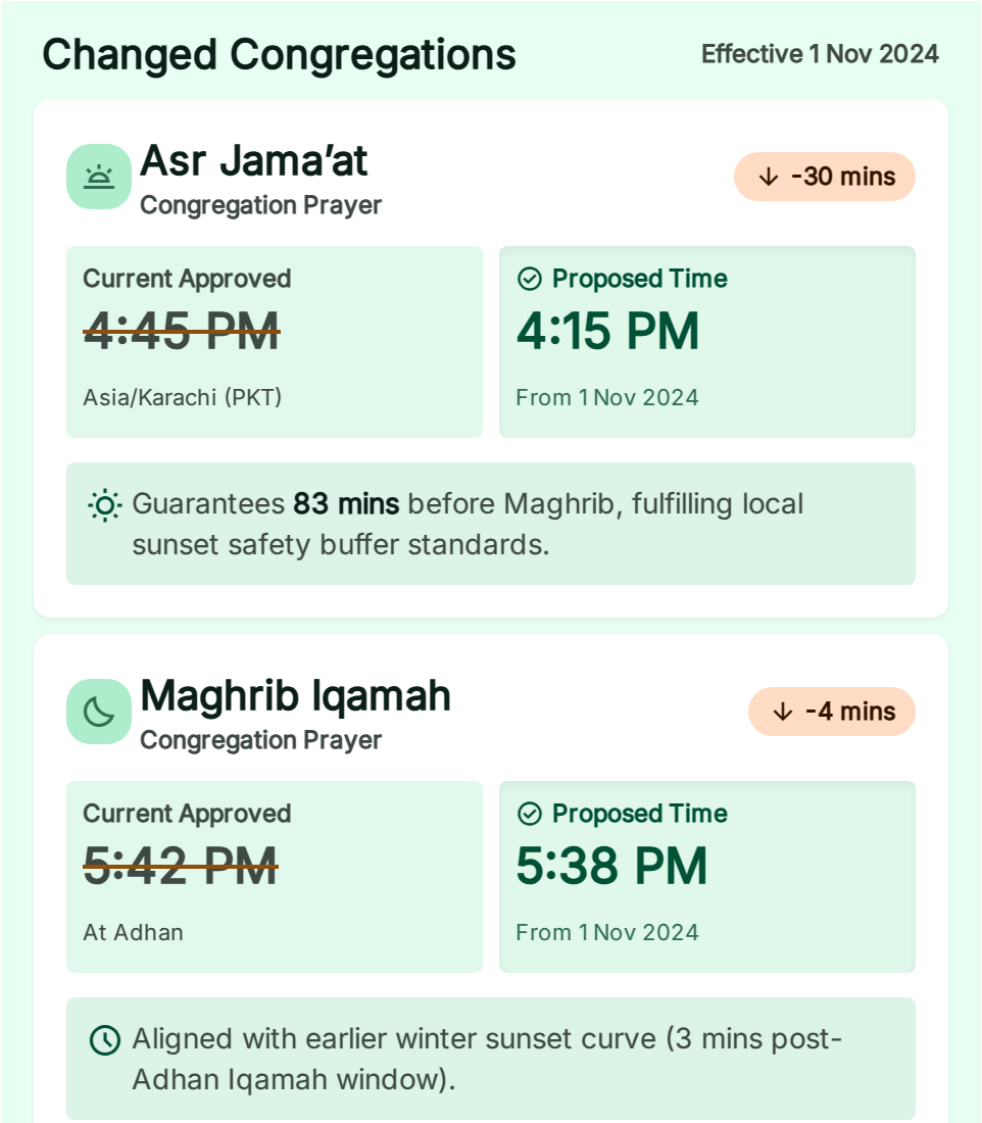


Figure 4.10 UI10 selected task region from the supplied prototype

Flow: Submit immutable snapshot once, then UI11 with version and review task reference.

Acceptance focus: Disable duplicate request while in flight; retry safely; conflict reloads comparison. No publication success claim here.

4.2.11 UI11 Submission Details and Status

Actor: Author and authorized reviewer. Read exact version, decision reason, activation status and history.

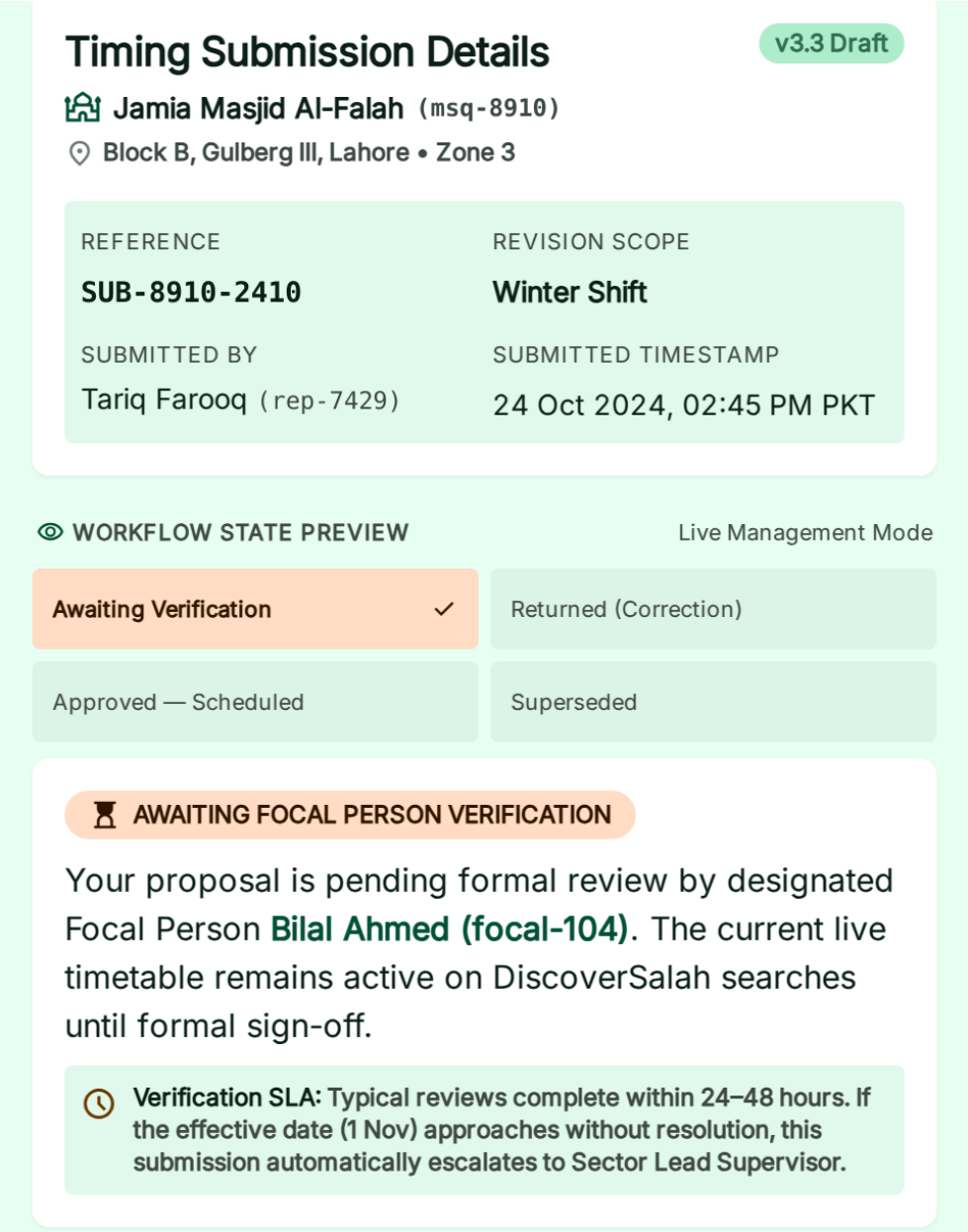


Figure 4.11 UI11 selected task region from the supplied prototype

Flow: Returned proposal can lead to a new UI09 draft; review links require current authority.

Acceptance focus: Approved for future activation differs from currently published. Superseded versions remain read-only.

4.2.12 UI12 Focal Person Dashboard and Review Queue

Actor: Focal Person. See assigned tasks, oldest pending work, affected dates and maintenance gaps.

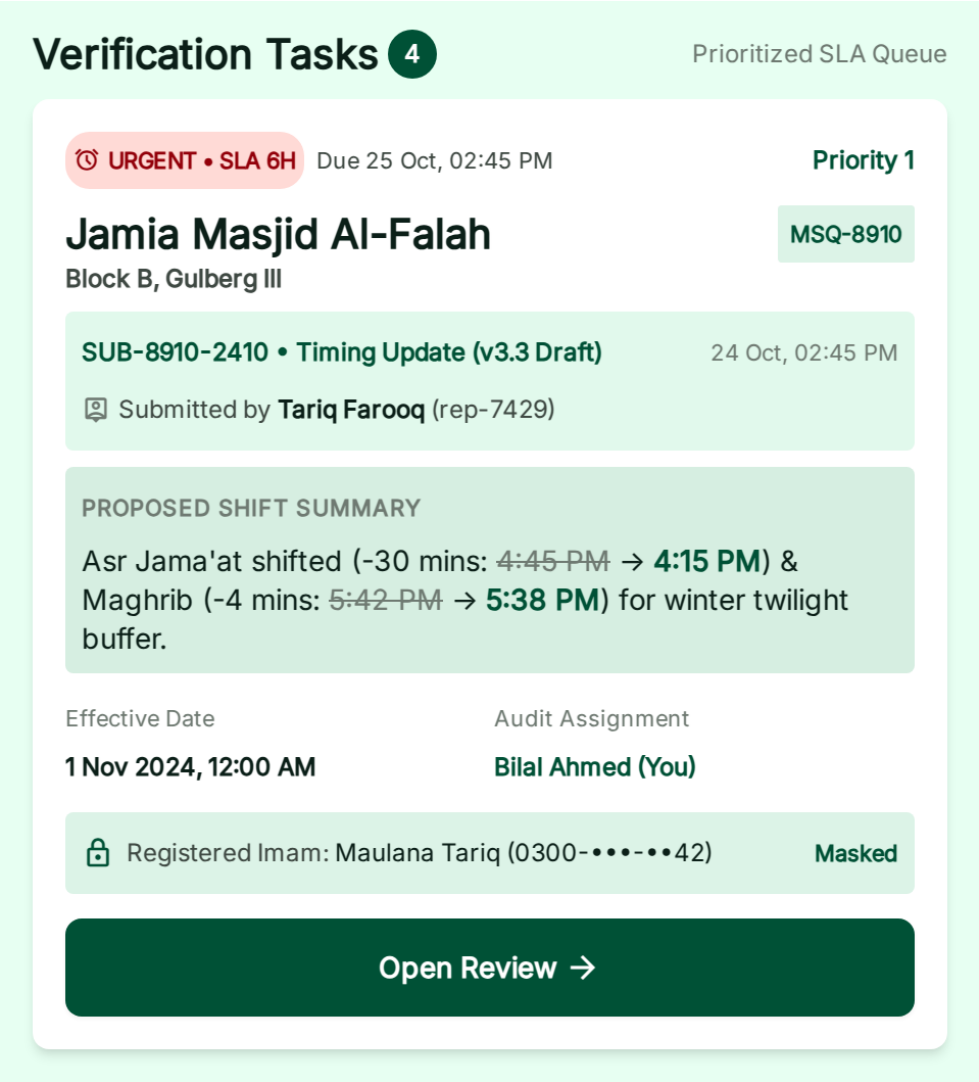


Figure 4.12 UI12 selected task region from the supplied prototype

Flow: Open UI13 for the current version; registration tasks open the appropriate verification form.

Acceptance focus: Queue exists even if notification delivery fails. Display aging honestly; no invented service-level promise.

4.2.13 UI13 Timing Verification and Approval

Actor: Independent scoped Focal Person. Compare changes, call the registered Imam, record evidence and decide exact version.

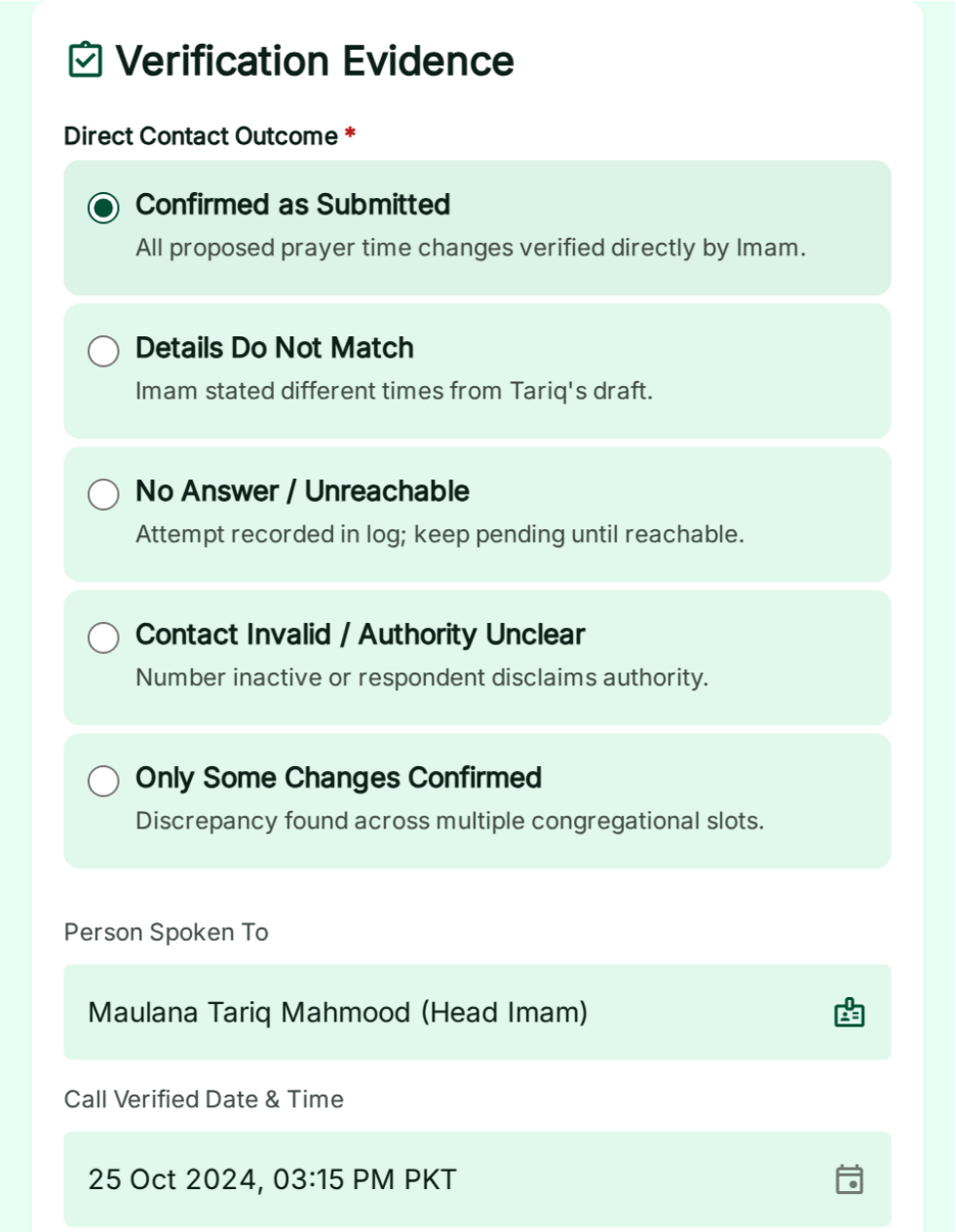


Figure 4.13 UI13 selected task region from the supplied prototype

Flow: Approve, return or reject with reason; UI11 records the decision and activation state.

Acceptance focus: Block self-approval, stale versions and incomplete evidence. Do not pre-check verification consent or call outcomes.

4.2.14 UI14 Mosque Registration

Actor: Authorized Local Representative. Capture mosque identity, entrance, private Imam contact permission, initial sessions and facilities.

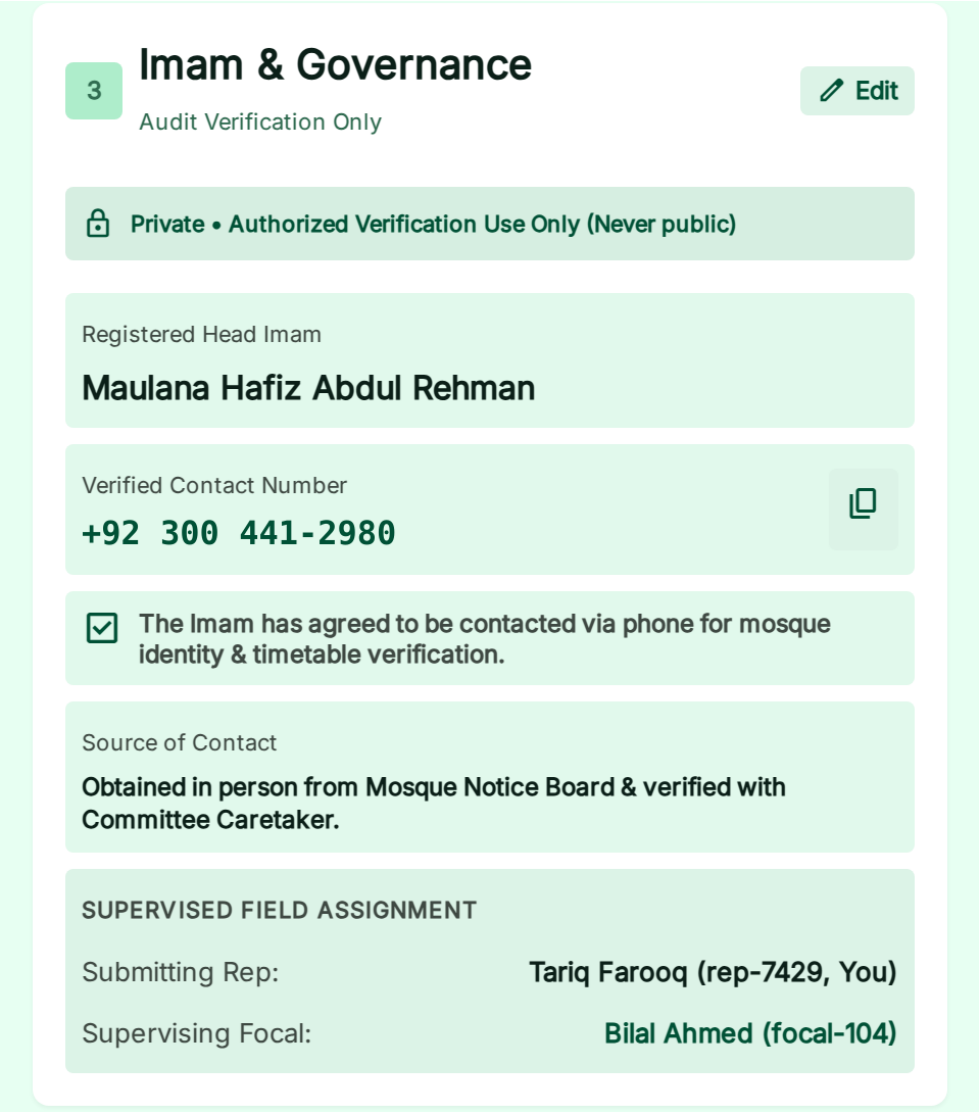


Figure 4.14 UI14 selected task region from the supplied prototype

Flow: Check duplicates, review registration and submit to Focal verification queue.

Acceptance focus: A local representative entering a number does not make it verified. Public activation requires independent decision.

4.2.15 UI15 Admin Dashboard and Assignment Management

Actor: Admin. Review uncovered areas and assignments, issue focal credentials, manage authorized credential resets/contact changes and maintenance responsibility. Representative credentials are issued only by the responsible focal person, not directly by Admin.

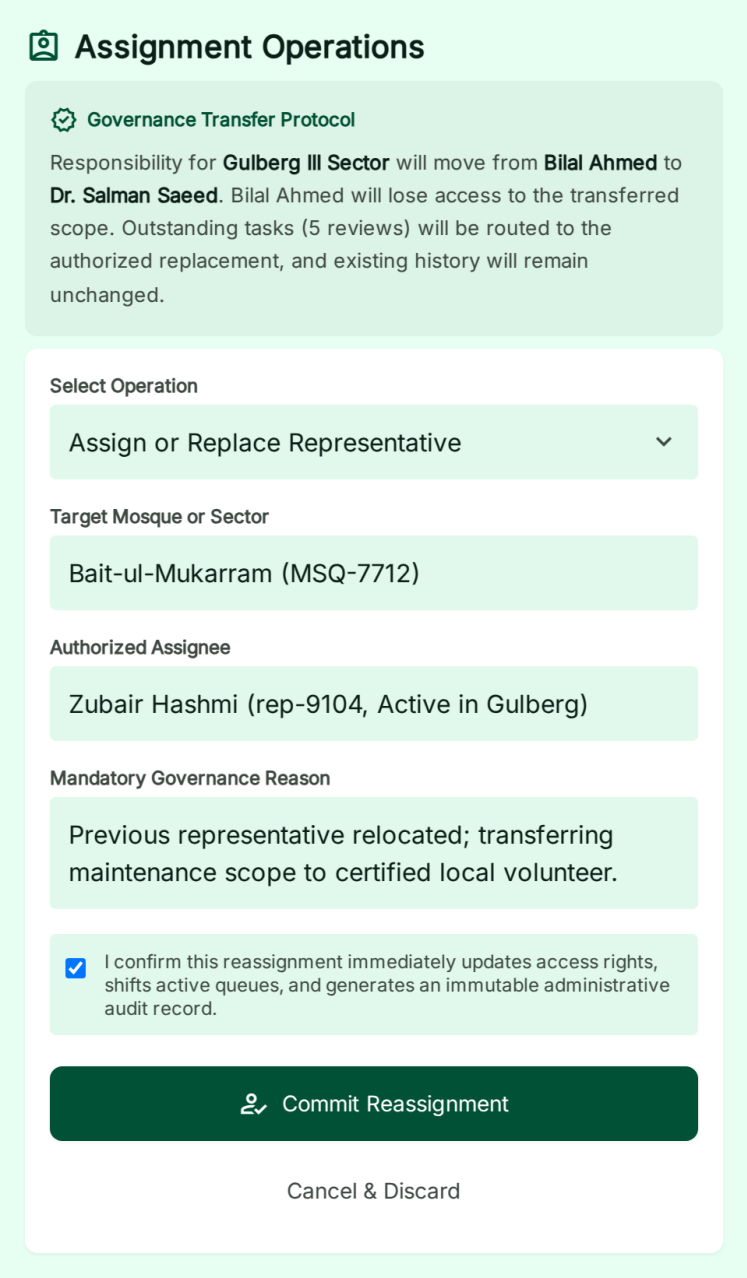


Figure 4.15 UI15 selected task region from the supplied prototype

Flow: Apply scoped reassignment with reason and review affected tasks; preserve audit history.

Acceptance focus: Revoke former writes immediately; no admin self-approval bypass. MFA badge must reflect real server state.

4.3 Prototype corrections and supporting coverage

These are inspection findings in the supplied HTML. They are open design/implementation tasks, not independently validated software defects. Do not adopt the conflicting text as a new requirement.

ID / severity	Pages	Correction required
PF01 / High	UI06	Anonymous timing updates are presented as permitted. Allow public reports/suggestions only; official proposals require authorized management.
PF02 / High	UI09/10/13	30-minute twilight/Sunnah rule and guaranteed separation appear without an approved policy. Remove religious inference and guarantee; capture Imam-confirmed times and actual change reason.
PF03 / High	UI05/08/13	Representative/committee/trustee labels blur the registered Imam verification workflow. Distinguish data entry, rep reconfirmation and Focal call verification; require documented exception policy before substitutes.
PF04 / High	UI07/15	30-day expiry appears as established policy. Use approved configurable freshness thresholds; retain proposed 7/14-day and event rules until decision.
PF05 / High	UI03/05	Driving advice and inconsistent travel/preparation buffers. Limit evaluated modes to walking/motorcycle; require validated uncertainty and transparent allowances.
PF06 / Medium	All pages	Viewport disables zoom; 44 px controls and token drift occur. Enable zoom, normalize tokens and use 48 px target; test keyboard/reflow/contrast.
PF07 / Medium	UI06	Forgot-password action invokes expired-session scenario. Implement verified recovery and proper validation/errors.
PF08 / Medium	Multiple	Hard-coded historical dates, counts, verification badges, 24–48 hour review promise and fixture data. Bind to server data/clock; approve service hours before promises; exclude fixtures from production.
PF09 / Medium	Multiple	Prototype links and scenario controls do not prove connected workflows. Build real routes and role guards; remove developer scenario panels from release.
PF10 / Medium	All pages	External asset dependencies can fail, leaving logo/icon fallbacks. Bundle approved assets and fonts; retain useful text alternatives and test failed asset requests.

4.4 Additional information

Supporting UI still required

Workflow or state	Placement	Acceptance requirement
Missing-mosque suggestion	Public result/detail supporting form	Reference and private tracking token; duplicate link; no automatic listing.
Contributor application	Public contribution form and admin review	Area, contact, availability and permission; verified invitation follows appointment.
Issue report and resolution	Public detail form and scoped management view	Report lifecycle and specific disputed scope; no automatic timetable rewrite.
Notice moderation	Management form and public venue notice board	Authority/consent checks for death/Janazah details, expiry, cancellation and no fundraising.
Account support	Invite acceptance, MFA, recovery and expired-session views	Token expiry, verified recovery, revocation and safe return location.
Special schedules	Editor subviews for Friday/Eid/dated exceptions	Multiple sessions, provisional dates, explicit cancellations and venue entrance.
Cross-cutting states	Within each core page	Loading, empty, denied, conflict, offline request failure and confirmation feedback.

Use these as focused forms, tabs or states where practical. The 15 main page designs remain the initial navigation set; they do not remove the supporting requirements from the release backlog. If capacity requires deferring a workflow, record the scope decision rather than marking it complete.

Screenshot handling

Rendered crops preserve supplied UI content and demonstrate selected task regions. The local supplied logo is used where the remote logo asset fails; fonts and icons are loaded for readable rendering. Illustration photos and map placeholders are not field evidence. The figures do not certify responsiveness or functional navigation. Do not store example personal details as production records.

5 Other design details

5.1 Security and privacy

Use opaque server sessions in Secure, HttpOnly, SameSite cookies with CSRF protection and origin checks for state-changing requests. Rate-limit authentication and public submissions; use safe generic login errors. Admin access requires MFA. Authorization evaluates role, active account, scope, assignment revision and no-self-approval on the server. Recovery and retention follow Section 5.1.1. Session and invitation expiry settings must be configured and tested before launch.

Allowlist public fields rather than deleting known private fields after serialization. Keep registered Imam numbers and verification notes available only to authorized management. Avoid logging passwords, tokens, precise origins and full phone numbers. Limit contact collection to purpose and permission; do not collect CNIC by default. Restrict audit access and make app-level history append-only.

5.1.1 Privacy retention and account recovery

Visitors can browse, search and navigate without signing in. Request location only when needed and offer manual entry. Use precise origin coordinates transiently for the query; do not create a location-history record or include coordinates in application logs. Explain the transfer to Google when requesting routes; provider processing is separate from our retention policy.

Store the Imam verification number privately with permission to contact. Permit access only to assigned personnel who need it and authorized administration. A public mosque office contact is a separate optional field with explicit publication permission. Changing private contact details requires independent evidence; a new number cannot verify itself.

Management accounts are individually issued. Only Admin creates focal credentials; only the responsible focal person creates representative credentials within assigned areas. Issuers confirm identity and contact details independently, then privately hand over a generated temporary password valid for 24 hours. The account owner must change it before managing records. No verified-email or SMS OTP onboarding/recovery channel is required.

Recovery follows the same issuer hierarchy. The issuer rechecks identity/contact before resetting credentials or changing the account contact number. These commands revoke sessions; temporary-password reset requires a new private password. Administrator sign-in uses locally verified TOTP with single-use offline recovery codes. Lost administrator MFA requires controlled operator recovery, never a public bypass.

Record	Retention and deletion rule
Unsuccessful or withdrawn applications	Delete personal contact details 90 days after rejection or withdrawal. Review unresolved applications at 90 days; do not retain them indefinitely without a documented purpose.
Security logs	Delete after 90 days; exclude passwords, tokens, private phone details and precise visitor coordinates.
Schedule changes and approval history	Retain for 12 months after the recorded action with minimal actor identifiers. Keep currently effective schedule and necessary provenance while active; do not delete the live timetable merely because it is old.
Backups	Encrypted daily backups, rolling 30-day retention, restricted access. Test restoration before launch. After restore, reapply deletion records before serving data.
Active contacts and other personal records	Keep only while needed for the active role or service. Remove obsolete contact details when their purpose ends. Review closed reports/notices for removal or anonymization; retain no unnecessary personal narrative.

Retention deletion is an authorized, auditable process distinct from routine append-only history. Preserve minimal deletion markers long enough to prevent backup restoration from reintroducing deleted data. These are accepted operational defaults, not a claim of statutory retention requirements.

5.2 Reliability and operations

Approval commits independently of notification delivery. The worker retries with leases, bounded backoff and deduplication; repeated failure raises an operational alert. A delayed activation worker cannot expose future timings early because reads check effective instants. A revoked or missing maintenance assignment makes the affected scope maintenance-pending and blocks positive arrival claims until coverage is restored.

Proposed operational targets are 99.5% availability, daily backups, RPO 24 hours and RTO 4 hours. Daily encrypted backups and a 30-day rolling backup retention are accepted. Availability, RPO and RTO remain planning targets until the separate database setup and restore rehearsal establish feasibility. Monitor API errors, route timeouts, oldest review age, stale coverage, outbox failures and assignment gaps; route initial alerts to the Product Owner and document a trusted emergency handover path before pilot.

5.3 Freshness and event governance

Proposed reconfirmation/freshness thresholds of 7 and 14 days and final event verification within 48 hours require owner approval. Store separate timestamps for representative reconfirmation and independent phone verification. A changed proposal or dispute affects its exact prayer/date scope, not every unaffected session. Expired or provisional event information must not remain positively ranked.

5.4 Announcements and growth

Moderated notices cover religious events, service information and death/Janazah announcements with source authority and consent appropriate to the content. Publish minimal necessary information, provide correction/cancellation and enforce expiry on reads. Janazah route feasibility is deferred unless separately specified. The team visits mosques to obtain permission and verify initial listings. After activation, use a management-approved poster with a website link and mosque-specific QR code, and Imam announcements. Social posts invite Focal applicants; review and scoped invitation precede access. No paid placement changes ranking.

6 Test specification and results

6.1 Test strategy and entry criteria

Testing begins with pure schedule and arrival rules, then API integration against a transaction-capable database, UI journeys and recovery/security checks. Provider contract tests use deterministic stubs; field journeys validate actual route uncertainty separately. No stub result establishes real-world route accuracy.

Entry requires a versioned build, approved test configuration, isolated staging, transaction-enabled database, synthetic accounts/venues and controlled provider responses. Record commit/build ID, environment, parameters, tester, execution time, evidence path and actual results for every run. Do not use real Imam phone numbers in automated fixtures.

Fixture	Definition
Accounts	R1 assigned representative; F1 independent Focal Person; A1 admin; R2 out of scope; revoked account R3.
Venues	V1 pilot area A; V2 separate area B; V3 maintenance-pending; private contact fixture never public.
Schedule	V1 head revision 12, approved S12; proposed S13. Asia/Karachi; frozen date 15 January 2027 solely for tests.
Arrival	Selected Dhuhr start 13:30; departure 13:00; stub route 15 min; prep 5; uncertainty 5. Boundary variants specified per case.
Failure controls	Provider timeout/no-route, dropped response after commit, paused worker, concurrent reviewer/revocation and delayed activation.

Traceability

Each test names its related SRS capability and Phase 3 section/UI. The SRS US/FR/AC suffixes below follow the Phase 2 traceability register; NFR indicates the named quality requirement. Keep these links synchronized when acceptance criteria change. Future-departure story 16 remains conditional and is not claimed as released.

Result convention

All planned application cases below are Not run. Actual results are deliberately unclaimed. Section 6.2 reports the completed source inspection separately. A test becomes Pass only with retained evidence; blocked or failed cases remain visible in the release decision.

6.1 Test cases P3 T01 and T02

Identifier	P3-T01 Location fallback
Related requirements	SRS US/FR/AC 09; SRS location selection; §3.2; UI02
Short description	Location fallback
Preconditions	Visitor; location permission denied.
Input data	Manual address and valid entrance pin.
Detailed steps	1 Deny permission. 2 Enter origin manually. 3 Confirm and search.
Expected results	Search uses confirmed origin; no forced login or invented GPS accuracy.
Postconditions	Selected origin retained; no precise origin in logs.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

Identifier	P3-T02 Prayer and date continuity
Related requirements	SRS US/FR/AC 09–11; SRS public discovery; §3.1.1; UI01–05
Short description	Prayer and date continuity
Preconditions	Approved fixtures for Asr and Maghrib.
Input data	Asr on fixture date; no remaining Asr.
Detailed steps	1 Search Asr. 2 Switch list/map. 3 Open and return from details.
Expected results	Prayer/date persist; empty Asr result does not silently switch to Maghrib/tomorrow.
Postconditions	Query state unchanged.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

6.1 Test cases P3 T03 and T04

Identifier	P3-T03 Whole City scope
Related requirements	SRS US/FR/AC 10; SRS Whole City search; §3.1.1; UI03
Short description	Whole City scope
Preconditions	Venues at 6 km and 12 km inside city; one outside.
Input data	Nearby maximum 7 km, then Whole City; origin outside city variant.
Detailed steps	1 Query Nearby. 2 Switch Whole City. 3 Repeat with outside origin.
Expected results	Nearby respects radius; Whole City includes city venues beyond 7 km and excludes other-city venues.
Postconditions	Scope shown explicitly.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

Identifier	P3-T04 Arrival boundary classification
Related requirements	SRS US/FR/AC 12–13; SRS reachability; §3.1.1; UI03
Short description	Arrival boundary classification
Preconditions	Eligible session at 13:30; validated fixture model.
Input data	Depart 13:00; prep 5, uncertainty 5; routes 20, 21, 30 minutes.
Detailed steps	1 Evaluate each route variant. 2 Inspect label and margin.
Expected results	20 => reachable at zero margin; 21 => tight; 30 => unlikely. Allowances visible.
Postconditions	No data mutation.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

6.1 Test cases P3 T05 and T06

Identifier	P3-T05 Unsupported motorcycle route
Related requirements	SRS US/FR/AC 12; SRS walking/motorcycle; §2.5; UI03
Short description	Unsupported motorcycle route
Preconditions	Only car or bicycle route available.
Input data	Motorcycle selected; no validated motorcycle model.
Detailed steps	1 Request evaluation. 2 Simulate car fallback response.
Expected results	Arrival unavailable; no relabeling car/bicycle as motorcycle and no positive claim.
Postconditions	Timetable remains accessible.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

Identifier	P3-T06 Trust eligibility precedence
Related requirements	SRS US/FR/AC 06, 13, 17; SRS freshness; §3.1.1 and §5.3
Short description	Trust eligibility precedence
Preconditions	Eligible baseline and one variant per invalid state.
Input data	Missing, expired, provisional, stale, disputed, affected pending, unmaintained.
Detailed steps	1 Set each state. 2 Return short route. 3 Evaluate.
Expected results	Each invalid state returns unavailable with reason despite a fast route; unaffected session remains independent.
Postconditions	No trust state automatically cleared.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

6.1 Test cases P3 T07 and T08

Identifier	P3-T07 Route failure and partial ranking
Related requirements	SRS US/FR/AC 10–13; SRS routing failure; §3.1.1; UI03/04
Short description	Route failure and partial ranking
Preconditions	Three candidate sessions; one timeout.
Input data	8-second configured timeout; partial batch.
Detailed steps	1 Query. 2 Fail one route. 3 Paginate and refresh.
Expected results	Failed estimate unavailable; no zero-duration fallback or claim of globally fastest city order. Stable base cursor.
Postconditions	Results identify evaluated scope.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

Identifier	P3-T08 Refresh and stale read race
Related requirements	SRS US/FR/AC 11–13; SRS live refresh; §3.1.1
Short description	Refresh and stale read race
Preconditions	Active results and controllable server clock.
Input data	Advance 61 seconds; change schedule head during routing.
Detailed steps	1 Background/resume page. 2 Advance clock. 3 Approve changed head before evaluation returns.
Expected results	Refresh on resume/interval; final revision check discards stale positive result.
Postconditions	Displayed version matches current eligibility.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

6.1 Test cases P3 T09 and T10

Identifier	P3-T09 Recurring exceptions and cancellation
Related requirements	SRS US/FR/AC 05, 15; SRS schedules; §3.1.1; UI09
Short description	Recurring exceptions and cancellation
Preconditions	Recurring Dhuhr plus dated override.
Input data	Explicit cancellation for fixture date.
Detailed steps	1 Resolve ordinary day. 2 Resolve override day. 3 Cancel override and resolve.
Expected results	Correct source chosen; explicit cancellation never resurrects recurrence for the canceled scope.
Postconditions	History retained.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

Identifier	P3-T10 Friday and Eid sessions
Related requirements	SRS US/FR/AC 15; SRS special congregations; §3.1.1; UI05/09
Short description	Friday and Eid sessions
Preconditions	Friday multiple sessions and provisional outdoor Eid.
Input data	Two Friday starts; separate Khutbah; provisional Eid date.
Detailed steps	1 Resolve Friday. 2 Inspect Dhuhr. 3 Evaluate Eid.
Expected results	Distinct Friday sessions; no unconfirmed ordinary Dhuhr; Khutbah separate; provisional Eid unavailable for reachability.
Postconditions	Outdoor entrance remains venue-specific.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

6.1 Test cases P3 T11 and T12

Identifier	P3-T11 Timezone and validation
Related requirements	SRS US/FR/AC 05, 11; SRS timetable correctness; §3.1; UI09
Short description	Timezone and validation
Preconditions	Device timezone differs from Asia/Karachi.
Input data	13:30 local; invalid 25:90; reversed effective interval.
Detailed steps	1 Save valid time. 2 Retry invalid values. 3 Read public session.
Expected results	Wall-clock meaning preserved; invalid values rejected with field errors and no partial save.
Postconditions	Approved record unchanged on invalid request.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

Identifier	P3-T12 Scoped authorization
Related requirements	SRS US/FR/AC 01–02, 21; SRS role access; §3.1 and §5.1; UI06–15
Short description	Scoped authorization
Preconditions	R1 has V1 only; R2 has V2; anonymous caller.
Input data	Direct write/read URLs for private V1 records.
Detailed steps	1 Call as anonymous. 2 Call as R2. 3 Call as R1.
Expected results	Anonymous/out-of-scope denied; correct scope permitted; client-supplied role ignored.
Postconditions	No unauthorized state or private-data disclosure.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

6.1 Test cases P3 T13 and T14

Identifier	P3-T13 Atomic idempotent submission
Related requirements	SRS US/FR/AC 05–07; SRS timing submission; §2.2; UI10/11
Short description	Atomic idempotent submission
Preconditions	R1, V1 head 12 and draft S13.
Input data	Key K1; payload P; response lost after commit.
Detailed steps	1 Submit. 2 Drop response. 3 Retry P/K1. 4 Retry changed P/K1.
Expected results	One version/task/outbox set; same result for retry; changed payload rejected.
Postconditions	Approved S12 preserved; pending S13 recorded once.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

Identifier	P3-T14 Supersede and concurrent edit
Related requirements	SRS US/FR/AC 05–06; SRS proposal versioning; §2.2; UI09–11
Short description	Supersede and concurrent edit
Preconditions	S13 pending; two editors loaded revision 13.
Input data	Two different new snapshots with expected revision 13.
Detailed steps	1 Submit first. 2 Submit second. 3 Inspect old review task.
Expected results	First creates new immutable version; second gets 409; S13 superseded and cannot be approved.
Postconditions	One active pending proposal.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

6.1 Test cases P3 T15 and T16

Identifier	P3-T15 Independent verified approval
Related requirements	SRS US/FR/AC 06; SRS Focal authority; §2.2.1; UI13
Short description	Independent verified approval
Preconditions	F1 scoped independent reviewer; S13 current.
Input data	Registered Imam contact, call time, scope and confirmed outcome.
Detailed steps	1 Record completed call evidence. 2 Approve exact revision. 3 Read status.
Expected results	Decision, head, audit and outbox commit together; activation follows effective date.
Postconditions	Evidence linked to immutable S13.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

Identifier	P3-T16 Block invalid verification
Related requirements	SRS US/FR/AC 06, 08; SRS remote verification; §2.2.1; UI13
Short description	Block invalid verification
Preconditions	Current task; author and independent reviewer sessions.
Input data	Self-approval; blank evidence; unreachable Imam; superseded version.
Detailed steps	1 Attempt approval for each variant. 2 Inspect history.
Expected results	All attempts blocked or remain pending/returned; admin role alone does not bypass independence.
Postconditions	No false publication.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

6.1 Test cases P3 T17 and T18

Identifier	P3-T17 Future activation
Related requirements	SRS US/FR/AC 06, 15; SRS publication timing; §2.2.1; UI11
Short description	Future activation
Preconditions	Approved future version, current S12; worker paused.
Input data	Activation T; reads at T−1 second and T.
Detailed steps	1 Approve. 2 Read before T. 3 Read at T with worker delayed.
Expected results	Before T current version persists; at T effective policy resolves new version without early publication.
Postconditions	Audit distinguishes approval and activation.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

Identifier	P3-T18 Durable notification recovery
Related requirements	SRS US/FR/AC 07–08; SRS in-account notifications; §2.2 and §5.2
Short description	Durable notification recovery
Preconditions	Committed task/outbox; worker stopped.
Input data	Restart and duplicate delivery attempt.
Detailed steps	1 Submit with worker down. 2 Open queue. 3 Restart twice.
Expected results	Task visible immediately; eventual single recipient notification; retries logged safely.
Postconditions	No task lost or duplicate decision.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

6.1 Test cases P3 T19 and T20

Identifier	P3-T19 Registration and private contact
Related requirements	SRS US/FR/AC 03–04; SRS mosque registration; §3.2; UI14
Short description	Registration and private contact
Preconditions	Authorized R1; duplicate V1 nearby.
Input data	New venue plus Imam contact permission; duplicate variant.
Detailed steps	1 Submit duplicate. 2 Submit distinct venue. 3 Read public API before review.
Expected results	Duplicate review required; new venue pending; private contact absent publicly; no automatic verification.
Postconditions	Focal registration task created.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

Identifier	P3-T20 Reassignment and revocation race
Related requirements	SRS US/FR/AC 01, 08, 19; SRS assignment governance; §3.1; UI15
Short description	Reassignment and revocation race
Preconditions	R1 assigned; pending reviews exist.
Input data	Admin reassigns while R1 write is paused before commit.
Detailed steps	1 Start write. 2 Revoke/reassign. 3 Resume write.
Expected results	Old write fails revision guard; tasks transfer to authorized replacement; history unchanged.
Postconditions	Uncovered scope maintenance-pending if no replacement.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

6.1 Test cases P3 T21 and T22

Identifier	P3-T21 Authentication and recovery
Related requirements	SRS US/FR/AC 01–02; SRS secure access; §5.1; UI06
Short description	Authentication and recovery
Preconditions	Invited admin; recovery channel configured.
Input data	Invalid login, expired invite, MFA challenge, session logout.
Detailed steps	1 Exercise each state. 2 Replay expired token/session.
Expected results	Generic failure, rate limit, single-use valid recovery; admin MFA enforced; logged-out session rejected.
Postconditions	No credentials in logs.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

Identifier	P3-T22 CSRF and private serialization
Related requirements	SRS US/FR/AC 01–02, 14; SRS privacy/security; §5.1
Short description	CSRF and private serialization
Preconditions	Authenticated browser; public endpoints.
Input data	Cross-origin write; public venue/search/errors; private phone fixture.
Detailed steps	1 Send forged write. 2 Inspect public responses/logs. 3 Submit script text.
Expected results	Forged write denied; phone/tokens/origin absent; untrusted text safely rendered.
Postconditions	No injection or unauthorized mutation.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

6.1 Test cases P3 T23 and T24

Identifier	P3-T23 Reports and scoped disputes
Related requirements	SRS US/FR/AC 18; SRS issue reporting; §5.3; supporting UI
Short description	Reports and scoped disputes
Preconditions	Published multi-prayer timetable.
Input data	Report Asr on one date.
Detailed steps	1 Submit report. 2 Reviewer marks disputed scope. 3 Resolve with evidence.
Expected results	Report alone does not rewrite timetable; only affected scope becomes ineligible; resolution audited.
Postconditions	Resolved/dismissed report retains history.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

Identifier	P3-T24 Suggestions and applications
Related requirements	SRS US/FR/AC 25–26; SRS community contribution; §4.4
Short description	Suggestions and applications
Preconditions	Anonymous visitor; reviewer accounts.
Input data	Missing venue duplicate; Focal application.
Detailed steps	1 Submit. 2 Use private tracking token. 3 Review application.
Expected results	Reference returned; no public contact leak; duplicate links existing venue; no automatic role grant.
Postconditions	Appointment requires authorized invitation.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

6.1 Test cases P3 T25 and T26

Identifier	P3-T25 Notice verification and expiry
Related requirements	SRS US/FR/AC 20; SRS announcements; §5.4
Short description	Notice verification and expiry
Preconditions	Scoped moderator; sensitive notice fixture.
Input data	Death/event notice; expiry T; cancellation.
Detailed steps	1 Try publish without source evidence. 2 Add evidence. 3 Read at expiry/cancel.
Expected results	Unverified sensitive notice blocked; minimal public fields; expired/canceled notice absent even if worker delayed.
Postconditions	Audit retained; no funding/Janazah arrival feature implied.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

Identifier	P3-T26 Accessible core journeys
Related requirements	SRS US/FR/AC 22; SRS usability/accessibility; §3.3; UI01–15
Short description	Accessible core journeys
Preconditions	Implemented responsive UI with bundled assets.
Input data	360/768/1280 CSS px; keyboard; 200% zoom.
Detailed steps	1 Complete discover/submit/review journeys. 2 Check focus, labels and reflow.
Expected results	No blocked actions, clipped required content or color-only status; 48 px target policy checked.
Postconditions	Record issues; WCAG claim requires full audit.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

6.1 Test cases P3 T27 and T28

Identifier	P3-T27 Representative user evaluation
Related requirements	SRS US/FR/AC 22; NFR usability; SRS usability; §1.3
Short description	Representative user evaluation
Preconditions	Five consenting target users; test data.
Input data	Find mosque/session, understand uncertainty, submit or review by role.
Detailed steps	1 Run moderated tasks. 2 Record success/errors without coaching.
Expected results	Proposed target: at least 4 of 5 complete core task; misunderstandings create fixes, not hidden exclusions.
Postconditions	Anonymized notes and retest evidence retained.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

Identifier	P3-T28 Load and provider degradation
Related requirements	SRS US/FR/AC NFR performance; SRS performance; §5.2
Short description	Load and provider degradation
Preconditions	Staging sized like proposed pilot.
Input data	1,000 venues, 5,000 sessions, 50 concurrent users for 15 min.
Detailed steps	1 Run measured workload. 2 Inject route delays. 3 Review percentiles.
Expected results	Proposed p95 non-routing response ≤2 seconds; bounded route timeout; no integrity failures.
Postconditions	Record hardware, workload and error rate; not adoption forecast.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

6.1 Test cases P3 T29 and T30

Identifier	P3-T29 Backup and recovery
Related requirements	SRS US/FR/AC 24; NFR recovery; SRS reliability; §5.2
Short description	Backup and recovery
Preconditions	Backup and isolated restore environment.
Input data	Known approved/pending versions, audit and outbox.
Detailed steps	1 Backup. 2 Simulate loss. 3 Restore. 4 Reconcile/replay worker.
Expected results	Proposed RPO ≤24h and RTO ≤4h demonstrated; no duplicate notifications or lost approved pointers.
Postconditions	Recovery report and cleaned test environment.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

Identifier	P3-T30 Field route calibration
Related requirements	SRS US/FR/AC 12–13, 23–24; SRS conservative reachability; §3.1.1
Short description	Field route calibration
Preconditions	Selected provider, consent, approved travel safety protocol.
Input data	30 journeys per mode on representative pilot routes, peak/off-peak and Friday.
Detailed steps	1 Record predictions/departure/arrival. 2 Analyze errors separately by mode. 3 Set/retest allowances.
Expected results	Proposed sample coverage ≥95% after allowances; inadequate mode remains timetable-only. Small sample is not a guarantee.
Postconditions	Evidence and approved uncertainty configuration versioned.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

6.1 Test cases P3 T31 and T32

Identifier	P3-T31 Core navigation and failures
Related requirements	SRS US/FR/AC 09–14, 21–22; SRS coherent UI; §4.1; UI01–15
Short description	Core navigation and failures
Preconditions	Integrated routes and permissions.
Input data	All core transitions; dropped connection on draft/submit/review.
Detailed steps	1 Follow each workflow. 2 Interrupt request. 3 Recover and revisit status.
Expected results	No dead links; retained drafts/query; server status resolves ambiguous commit before retry.
Postconditions	No duplicate write or false success banner.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

Identifier	P3-T32 Release scope and readiness
Related requirements	SRS US/FR/AC 23–24; SRS pilot scope; §7
Short description	Release scope and readiness
Preconditions	Candidate build and evidence register.
Input data	Open defects, assignments, policies and feature flags.
Detailed steps	1 Review gates. 2 Inspect public copy/features. 3 Approve or hold rollout.
Expected results	No subscriptions/ads/payroll; no unsupported mode claims; named coverage; high-risk defects closed with evidence.
Postconditions	Recorded owner decision; hold if gate unmet.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

6.1 Test cases P3 T33 and T34

Identifier	P3-T33 Unchanged confirmation preserves trust flags
Related requirements	SRS US/FR/AC 17; SRS unchanged reconfirmation; §5.3; UI07/08
Short description	Unchanged confirmation preserves trust flags
Preconditions	Pending/disputed Asr; last Focal phone date fixed.
Input data	Representative confirms no change.
Detailed steps	1 Reconfirm. 2 Inspect phone timestamp. 3 Evaluate disputed session.
Expected results	Rep reconfirmation timestamp changes; phone timestamp and pending/dispute flags do not clear.
Postconditions	Affected session remains ineligible.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

Identifier	P3-T34 Private contact replacement
Related requirements	SRS US/FR/AC 19; SRS contact governance; §5.1; UI08/14
Short description	Private contact replacement
Preconditions	Current registered Imam number; new number submitted by R1.
Input data	Replacement contact and reason.
Detailed steps	1 Propose new contact. 2 Attempt approval solely through that number. 3 Review independently.
Expected results	No automatic replacement or self-verification; trusted existing channel/independent evidence required before approval.
Postconditions	Old contact remains controlled; decision audited.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

6.1 Test cases P3 T35 and T36

Identifier	P3-T35 Concurrent review decisions
Related requirements	SRS US/FR/AC 06; SRS exact-version approval; §2.2.1; UI13
Short description	Concurrent review decisions
Preconditions	Two authorized independent reviewers opened S13 at same revision.
Input data	One approve and one return command released simultaneously.
Detailed steps	1 Record valid evidence. 2 Send both commands. 3 Inspect head and audit.
Expected results	Exactly one decision commits; other returns conflict and must reload; no contradictory status.
Postconditions	Single authoritative decision and notification event.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

Identifier	P3-T36 Temporary backup scope expires
Related requirements	SRS US/FR/AC 08; SRS delegated review; §5.2; UI12/13
Short description	Temporary backup scope expires
Preconditions	Backup Focal assignment with start/end instants.
Input data	Task in scope; command before start, inside window and after expiry.
Detailed steps	1 Execute each time variant. 2 Repeat on other area.
Expected results	Only in-window, in-scope request allowed; expiry checked at commit; queued tasks remain owned.
Postconditions	No lingering backup privilege.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

6.1 Test cases P3 T37 and T38

Identifier	P3-T37 Bilingual continuity
Related requirements	SRS US/FR/AC 22; Decision 15 Oct 2026; §3.4; UI01–15
Short description	Bilingual continuity
Preconditions	Both locales implemented; public query and unsaved management form.
Input data	English/Urdu; mixed names, times and numbers.
Detailed steps	1 Switch language on search. 2 Switch with unsaved form. 3 Check RTL, errors and keyboard focus.
Expected results	Route, filters and entered data preserved; translated core labels; no clipped Urdu or reversed numeric values.
Postconditions	Preference retained without changing authorization or timetable values.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

Identifier	P3-T38 Recovery and token expiry
Related requirements	SRS US/FR/AC 01–02; Decision 15 Oct 2026; §5.1.1; UI06
Short description	Recovery and token expiry
Preconditions	Verified email, active sessions and admin MFA recovery fixture.
Input data	Valid token; reused token; token at 15-minute expiry; unknown email.
Detailed steps	1 Request reset. 2 Reset once. 3 Retry token and expired variant. 4 Check sessions and notification.
Expected results	Generic request response; expired/reused token rejected; sessions revoked after success; admin recovery does not bypass MFA identity checks.
Postconditions	Audit and notification retained without storing plaintext tokens.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

6.1 Test cases P3 T39 and T40

Identifier	P3-T39 Retention and restored deletion
Related requirements	SRS US/FR/AC NFR privacy and recovery; Decision 15 Oct 2026; §5.1.1
Short description	Retention and restored deletion
Preconditions	Clock-controlled records and isolated backup restore.
Input data	Applications/logs 90 days; history 12 months; backups 30 days.
Detailed steps	1 Run retention at boundaries. 2 Check active schedule. 3 Restore older backup. 4 Reapply deletions before access.
Expected results	Expired personal records removed; current schedule preserved; backups rotate; deleted contacts not reintroduced into service.
Postconditions	Minimal deletion audit and restore evidence recorded.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

Identifier	P3-T40 Mosque launch and route limits
Related requirements	SRS US/FR/AC 03–04, 12–13, 23–24; Decision 15 Oct 2026; §3.4 and §7.4
Short description	Mosque launch and route limits
Preconditions	Single consenting mosque; independent Focal; route quota fixture.
Input data	Verified entrance, permission, QR destination and exhausted route quota.
Detailed steps	1 Activate after verification. 2 Open QR. 3 Search both modes. 4 Exhaust quota.
Expected results	One mosque can launch; QR opens correct listing; no unsupported positive estimates; timetable and unrouted city results remain available.
Postconditions	Account approval remains controlled; usage-limit incident visible to owner.
Actual results	Not run; build and execution evidence pending.
Test case result	Not run

6.2 Summary of test results

No application test executions have been evidenced for this phase. Zero runs below is a coverage status, not proof of quality. Defect counts from unexecuted testing are unknown. The prototype findings are tracked separately to avoid presenting a static inspection as software validation.

Module	Cases planned	Run	Found / fixed / open
Discovery and schedules	T01–T11	0	Unknown / not evidenced / unknown
Access and publication	T12–T22	0	Unknown / not evidenced / unknown
Community workflows	T23–T25	0	Unknown / not evidenced / unknown
Quality and release	T26–T32	0	Unknown / not evidenced / unknown
Additional trust controls	T33–T36	0	Unknown / not evidenced / unknown
Accepted launch policies	T37–T40	0	Unknown / not evidenced / unknown
Complete system	40 case specifications	0	No execution-based defect total available

Completed prototype inspection

Review activity	Observed result
Source inventory	15 HTML screen prototypes available; 8 PNG placeholders failed; 7 previews too narrow for print.
Static source review	10 grouped findings PF01–PF10; all remain open against product source. Five High and five Medium.
Figure preparation	Readable task-region crops produced from HTML with local rendering assets.
Functional validation	Not established by screenshots, HTML links or hard-coded success states.
Corrections verified in product	None evidenced; documentation records required changes without claiming source fixes.

Defect triage and evidence record

High covers timing trust, authorization, privacy or arrival claims that can materially mislead users. Medium covers impaired workflow, accessibility and consistency. Engineering owns correction; QA verifies the same build with regression evidence; the Product Owner accepts remaining lower-risk limitations. Record ID, severity, affected version, reproduction, expected/actual result, owner, status, evidence and verified fix build.

6.3 Release acceptance gates

Gate	Evidence required	Decision owner
Timing integrity	T04–T11, T13–T20 and T33–T36 pass; no high-risk timing/authority defect remains.	Engineering and QA; Product Owner release decision
Route confidence	Google supports selected local routes; T30 evidence and approved allowances, or unavailable estimates for unsupported modes.	Engineering and local operations
Security and privacy	Authorization, MFA, CSRF, recovery and private-data review; T38–T39 evidence.	Engineering and QA
Usability	Core navigation, English/Urdu and accessibility results; T37 evidence and observed user tasks.	UI/UX and QA
Operational coverage	Named staff/backup, review hours, stale-data queue, incident owner and restore rehearsal.	Local operations and admin
Scope and documentation	Approved policy decisions; corrected prototype text; supporting forms implemented or formally deferred.	Product Owner

A critical or High unresolved issue in timing integrity, unauthorized access or private-data exposure blocks pilot release. Disabling an affected feature is acceptable only when the remaining experience is truthful and the Product Owner records the reduced scope. Passing tests does not establish universal prayer attendance or route guarantees.

Test report completion fields

For each executed case append build ID, environment, configuration version, tester, timestamp, actual result, status and evidence reference. Link any defect and rerun evidence. Retain provider-stub results separately from field observations. Record blocked tests and their dependencies rather than marking them passed by review.

7 Revised project plan

7.1 Current completion status

Module or work product	Status	Remaining work
Proposal and requirements	Documented baseline	Resolve policy decisions and approve changes.
Phase 2 architecture/design	Documented baseline	Validate during implementation; keep contracts and diagrams synchronized.
15 core UI pages	Partially implemented as standalone HTML prototypes	Fix PF01–PF10; connect routes and real data; supporting states/forms.
API, database and worker	Not evidenced as implemented	Build modules, migrations/indexes, transactions and operations.
Provider integration	Not evidenced as implemented	Configure Google provider; validate modes, terms, coverage and budget.
Automated and acceptance tests	Specified, not executed	Implement harness and execute retained test cases.
Field pilot and independent review	Not completed	Staff areas, calibrate, conduct review and record release decision.
Complete system	Not production-ready on current evidence	Complete release gates before public pilot.

The plan below retains eight candidate two-week sprint buckets from the Phase 2 planning baseline. They are sequencing estimates, not committed calendar dates. Team size, working hours and remaining supporting scope must be estimated before dates are promised. Testing occurs throughout, not only in the last sprint.

7.2 Conditional Gantt plan

The editable Word matrix is a relative Gantt chart. Filled cells indicate planned work windows; no cells imply completed implementation. Dependencies take precedence over overlapping cells.

Workstream	S1	S2	S3	S4	S5	S6	S7	S8
Decisions and prototype fixes	■■							
Foundation and identity	■■	■■						
Registry and registration		■■	■■					
Schedule and approval			■■	■■				
Discovery and route integration				■■	■■			
Reports applications notices					■■	■■		
Hardening and recovery						■■	■■	
Field evidence and pilot gate							■■	■■

7.3 Dependencies and delivery controls

Milestone	Exit evidence	Dependency
M1 Foundation	First consenting mosque, assigned maintainers, backlog, schema and secure sign-in.	Staff capacity and policy decisions.
M2 Trusted maintenance	Registration, proposal, exact-version review, audit/outbox and assignment tests.	M1 and actual operating workflow.
M3 Public discovery	Real timetable search, list/map, Whole City and honest route fallback.	M2; selected provider and query/ranking contract.
M4 Supporting operations	Reports, suggestions, applications, notices and accessible failure states.	M2; estimate supporting UI rather than hiding scope.
M5 Pilot ready	Security/load/restore/usability evidence, route calibration and staffed coverage.	M3/M4; resolved High findings and owner approval.

Remaining implementation prerequisites

Item	Required before
First mosque, consent record, staff identities and review cover	Activate each mosque only after verification and independent reviewer assignment.
Freshness warning after 7 days; 14-day restriction and event 48-hour rule remain provisional	Confirm exact timestamp basis and boundary behavior before release.
Google billing limit, request quotas and field-tested uncertainty	Enable positive walking or motorcycle estimates.
Separate MongoDB hosting, backup storage and durable job runner	Production deployment and restore rehearsal.
Verified-email service, session and invitation expiry configuration	Management onboarding and recovery testing.
Supporting forms, team capacity and calendar estimates	Commit release dates; do not infer completion from prototypes.

Track delivery by completed acceptance evidence and a working increment. Do not report sprint velocity, completion percentage or defect-removal time until measured. If route validation or staffing slips, offer truthful timetable-only coverage or delay the affected area; do not relax verification to meet a date.

7.4 Accepted launch decisions

Version 1.1 records the Product Owner's instructions and acceptance on 15 October 2026. These changes supersede conflicting pilot, language, provider and privacy assumptions in earlier documents. Synchronize the SRS and SDS during the next documentation update; this edition does not claim those files have already been revised.

Decision	Accepted implementation baseline
Mosque-led pilot	Any consenting mosque may start; management permission, an assigned maintainer, a registered Imam contact and an independent Focal reviewer are required. No minimum of 2–3 complete areas.
Recruitment and promotion	Founder/team visits mosques. Social posts recruit candidates; verify before invitations. Publish the mosque-specific QR poster only after the listing is active.
Languages	English/Urdu switching at launch; Urdu right-to-left layout; preserve route, query and form data when switching.
Reminders and notifications	No automated user reminders initially. In-account review and decision notifications remain essential. Recovery/invitation email remains transactional.
Maps	Google Maps and Routes API; WALK and TWO_WHEELER; accurate entrance pins, local validation and honest fallback.
Privacy and recovery	No public browsing account; private Imam verification number; separately authorized public office contact; invitation-only accounts and controlled recovery.
Operations	Vercel website hosting. Product Owner initially handles monitoring, support, backups and restoration, with documented emergency handover.
Retention	90 days for unsuccessful/withdrawn application contact data and security logs; 12 months for change/approval history; 30-day rolling encrypted backups.

Freshness thresholds remain provisional as explicitly stated in the preceding discussion. No approval of a maps provider constitutes proof of local routing accuracy. Database hosting, budget amounts and staffing names remain implementation prerequisites, not invented commitments.

References

[1] DiscoverSalah Project Proposal, version 4.0, 30 September 2026. Project scope and free pilot operating model.

[2] DiscoverSalah Phase 1 Software Requirements Specification, version 1.2, 4 October 2026. Functional scope, proposed quality targets and decision log.

[3] DiscoverSalah Phase 2 Agile Software Design Specification, version 1.1, 5 October 2026. Architecture, versioning, data relationships and source diagrams retained in Figures 2.1–2.5.

[4] Template DiscoverSalah DTS Phase 3. Supplied Phase III Design and Test Specification template. Section structure, test case fields, results, revised plan and IV&V appendix.

[5] DiscoverSalah screens.zip. Supplied 15 core HTML designs, preview images, logo and Devout Utilitarian DESIGN.md. Prototype source inspection and cropped interface figures. Reviewed 14 October 2026.

[6] MongoDB Documentation. Production Considerations for Transactions. https://www.mongodb.com/docs/manual/core/transactions-production-consideration/ Accessed 14 October 2026. Supports the transaction-capable replica-set deployment requirement in Section 2.4.

[7] World Wide Web Consortium. Web Content Accessibility Guidelines WCAG 2.2, Recommendation, 12 December 2024. https://www.w3.org/TR/2024/REC-WCAG22-20241212/ Accessed 14 October 2026. Accessibility target and text-resizing reference for Sections 3.3 and 6.1; no conformance certification is claimed.

[8] Google Routes API. Two-wheeler routing and supported countries. https://developers.google.com/maps/documentation/routes/route_two_wheel and https://developers.google.com/maps/documentation/routes/coverage-two-wheeled. Checked 15 October 2026; Pakistan listed; local validation remains required.

[9] OWASP Forgot Password and Multifactor Authentication Cheat Sheets. https://cheatsheetseries.owasp.org/cheatsheets/Forgot_Password_Cheat_Sheet.html and https://cheatsheetseries.owasp.org/cheatsheets/Multifactor_Authentication_Cheat_Sheet.html. Checked 15 October 2026.

[10] Vercel Storage overview. https://vercel.com/docs/storage. Checked 15 October 2026; database service and backup configuration remain separate deployment decisions.

Reference precedence and change control

The 15 October 2026 owner decisions recorded in Section 7.4 govern changed scope in this revision. Earlier approved requirements govern unaffected behavior. Phase 2 supplies the design baseline; synchronize its hosting and policy assumptions before implementation. Supplied prototypes illustrate presentation and may contain conflicting example text; those differences are handled in PF01–PF10. Third-party references explain technical constraints and do not establish project implementation results.

Appendix A Glossary

Term	Meaning
Jama’at	The mosque’s congregation session and stated start time, distinct from a calculated prayer window.
Focal Person	Appointed scoped reviewer who verifies information by contacting the registered Imam and decides submitted versions.
Local Representative	Authorized contributor responsible for assigned mosque records; cannot approve their own submission.
Registered Imam contact	Private contact captured with permission during registration for verification.
ScheduleHead	Mutable reference record holding approved/pending versions and concurrency revision.
Immutable proposal	Submitted snapshot whose content does not change; correction creates a new version.
Effective date	Date/instant when approved information is applicable to public resolution.
Reconfirmation	Representative statement that information is unchanged; separate from independent phone verification.
Reachability	Conditional arrival estimate using eligible timing, route duration, preparation and uncertainty.
Whole City	Search scope across participating venues in the selected city, without Nearby’s radius cap.
Outbox	Durable event record written atomically with domain changes for later worker processing.
Idempotency	Repeating the same command safely without creating duplicate effects.
RPO and RTO	Maximum targeted data-loss interval and restoration duration.
IV and V	Independent verification and validation by a reviewer who did not produce the assessed work.
Prototype	Interface demonstration; not evidence that backend behavior or security exists.

Appendix B Independent verification and validation

Review status

Independent verification and validation has not been performed for this edition. The source inspection documented here is part of preparation and must not be represented as independent assurance. An independent reviewer and review date remain to be appointed; no signature or endorsement is implied.

IV and V resource	Record
Reviewer name and organization	To be appointed by Product Owner.
Independence	Reviewer must not have authored the evaluated design or implementation.
Review baseline	DTS 1.1, SRS v1.2, SDS v1.1, candidate build and test evidence when available.
Review date and signature	Pending independent review; no signature recorded.
Outcome	Not assessed.

Required review procedure

Trace core requirements to interfaces and tests; inspect role/privacy boundaries; challenge arrival and freshness rules; replay approval/revocation races; sample actual test evidence; inspect readable screens and supporting workflows. Record non-trivial findings with severity and reproduction. Engineering corrects them and the independent reviewer verifies closure before issuing an outcome.

ID	Non-trivial defect	Origin	Status	Fix time
Pending	No independent findings recorded because the review has not occurred.	Not assessed	Not assessed	Not measured

PF01–PF10 are preparation-stage prototype findings in Section 4.3. They are available to the independent reviewer but are not substituted for an IV&V report. The final signed report should state reviewed versions, evidence, limitations, unresolved findings and recommendation.